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Semiconductor Electronics

SOLUTIONS

BEGINNER'S BOX-1

1. On heating resistance of semiconductor will decrease. So total resistance of the circuit decreases.
To keep reading of ammeter(A) constant value of R should be increase.

2. By mass-action law -

$n_i^2 = n_e \times n_h$

given $n_i = 1.5 \times 10^{16}/m^3$ $\quad n_h = 4.5 \times 10^{22}/m^3$

$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5\times10^{16})^2}{4.5\times10^{22}} = 5 \times 10^9/m^3$$

3. $N_D = 5 \times 10^{22}$ atoms/m^3 [1 ppm = 10^{-6}per atom]
for n–type semiconductor $n_e \approx N_D$

so $\boxed{n_e = 5\times10^{22}\ /m^3}$

By mass action law $n_i^2 = n_e \times n_h$

$$n_h = \frac{n_i^2}{n_e} = \frac{(1.5\times10^{16})^2}{5\times10^{22}} = 4.5\times10^9\ /m^3$$

4. We know that

$I_e = n_e A e V_e$ ...(1)

$I_h = n_h A e V_h$ ...(2)

Given $\frac{I_e}{I_h} = \frac{3}{1}$ and $\frac{V_e}{V_h} = \frac{5}{2}$

from eq (1) and (2) $\quad \frac{I_e}{I_h} = \frac{V_e}{V_h}.\frac{n_e}{n_h}$

$$\frac{3}{1} = \frac{5}{2}\times\frac{n_e}{n_h} \Rightarrow \boxed{\frac{n_e}{n_h} = \frac{6}{5}}$$

BEGINNER'S BOX-2

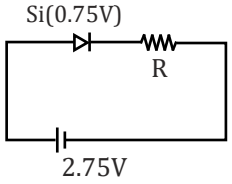
1. Potential barrier across the junction = 0.2 V

(a) When junction unbiased then K.E. required to cross the junction = 0.2 eV

(b) When junction is FB with 0.1 V then effective barrier potential = 0.2–0.1 = 0.1V
so required KE = 0.1 eV

(c) When junction is RB with 0.1 V then effective barrier potential = 0.2+0.1= 0.3V
so required KE = 0.3 eV

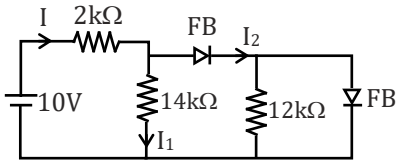
2.



$$\frac{(2.75-0.75)V}{R} = 10mA$$

$R = 200\Omega$

3.



$$I = I_2 = \frac{10V}{2k\Omega} = 5\ mA,\ I_1 = 0$$

4. $$I = \frac{(1.7-0.7)V}{2+2} = \frac{1V}{4\Omega} = 0.25A$$

5. (i) When both diode are in FB then -

$V_0 = (12–0.3)V = 11.7\ V$

$$I = \frac{V_0}{R} = \frac{11.7V}{5k\Omega} = 2.34\ mA$$

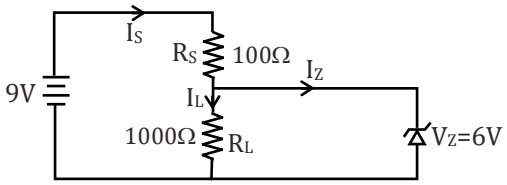
(ii) When Ge diode in RB then -

$V_0 = (12 – 0.7)V = 11.3V$

$$I = \frac{11.3V}{5k\Omega} = 2.26\ mA$$

BEGINNER'S BOX-3

2.



$V_Z = I_L R_L$ $\quad 9 – 6 = I \times 100$

$6 = I_L \times 1000$ $\quad I_s = 30mA$

$I_L = 6mA$ $\quad I_Z = I_s – I_L = 24mA$

$P_Z = I_Z.V_Z = (24\ mA)\ (6V) = 0.144\ W$

3. given that $V_Z = 9.1\ V$

$P_{max} = 364\ mW$

$P_{max} = V_z \times (I_z)_{max}$

$$(I_z)_{max} = \frac{P_{max}}{V_z} = \frac{364mW}{9.1V} = 40mA$$

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4. energy of emitted photons $\le \Delta E_g$

$$\frac{hc}{\lambda} \le \Delta E_g$$

$$\lambda \ge \frac{hc}{\Delta E_g} \ge \frac{12400}{\Delta E_g(eV)}$$

$$\lambda \ge \frac{12400}{1.43} \ge 8671.3\text{Å} \Rightarrow \lambda_{min} = 8671.3\ \text{Å}$$

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Modern Physics

SOLUTIONS

Modern Physics-1 [Dual Nature of Radiation and Matter]

BEGINNER'S BOX-1

1. Hertz

2. $E = 3keV = 3000\ eV$

$= 3000 \times 1.6 \times 10^{-19}\ J$

$= 4800 \times 10^{-19}\ J = 48 \times 10^{-17}\ J$

$p = \frac{E}{c} = \frac{48\times10^{-17}}{3\times10^{8}} = 16 \times 10^{-25}$

$p = 1.6 \times 10^{-24}\ kgm\text{-}s^{-1}$

3. $\lambda = 500\ nm = 500 \times 10^{-9}\ m = 5 \times 10^{-7}\ m$

$P = 10\ W$

(a) $n = 5 \times 10^{24}\ P\lambda \quad \left(P = n\frac{hc}{\lambda}\right)$

$n = 5 \times 10^{24} \times 10 \times 5 \times 10^{-7}$

$n = 25 \times 10^{18}$ photons/sec

$n = 2.5 \times 10^{19}$ photons/sec

(b) for perfectly absorbing surface -

$F = \frac{P}{c} = \frac{10}{3\times10^{8}} = 3.3 \times 10^{-8}\ N$

4. $E = \frac{hc}{\lambda} \quad \Rightarrow \quad E \propto \frac{1}{\lambda}$

$\lambda_{violet} < \lambda_{Red}$

$E_{violet} > E_{Red}$

5. $P = 100\ MW = 100 \times 10^{6}\ W = 10^{8}\ W$

$\nu = 10\ MHz = 10^{7}\ Hz$

$\lambda = \frac{c}{\nu} = \frac{3\times10^{8}}{10^{7}} = 30\ m$

$n = 5 \times 10^{24}\ P\lambda$

$n = 5 \times 10^{24} \times 10^{8} \times 30$

$n = 15 \times 10^{33}$ photons/sec

$n = 1.5 \times 10^{34}$ photons/sec

6. $I = 100\ W/m^2$

$\lambda = 400\ nm$

$I = \frac{n}{A}\frac{hc}{\lambda}$

$\frac{n}{A}$ = no. of photons per unit area per unit time

$\frac{n}{A} = \frac{I\lambda}{hc} = \frac{100\times400\times10^{-9}}{6.6\times10^{-34}\times3\times10^{8}}$

$\frac{n}{A} = 2.02 \times 10^{20}$

7. $\lambda = 700\ nm$

$P = 60\ W$

$\eta = 50\%$

$\because$ Bulb efficiency in converting electrical energy to light energy is 50%

$\because$ Power radiated by bulb $= \frac{50}{100}\times60 = 30W$

No. of photons emitted per sec = n

$n = 5 \times 10^{24} \times 700 \times 10^{-9} \times 30$

$n = 105 \times 10^{18}$ photons/sec

Number of photons emitted in 1 day is

$= 105 \times 10^{18} \times 3600 \times 24$

$= 9.072 \times 10^{24}$

BEGINNER'S BOX-2

1. $\phi = 4eV$

$\lambda = 5000\ Å$

$E = \frac{12400}{5000} = 2.48\ eV$

$\because$ Energy of photon $< \phi$

$\therefore$ photo electric effect does not occur.

2. $i_1 = 4$ unit $\quad d_2 = 1$ unit

$d_1 = 4$ unit $\quad i_2 = ?$

$i \propto \frac{1}{d^2}$

$\frac{i_2}{i_1} = \frac{d_1^2}{d_2^2} = \left(\frac{4}{1}\right)^2 = 16$

$i_2 = 16 i_1 = 64$ unit

3. $\lambda_1 = 4000\ Å \qquad \lambda_2 = 3600\ Å$

$E_1 = \frac{12400}{4000} = 3.1\ eV \qquad E_2 = 3.44\ eV$

$eV_{01} = E_1 - \phi \qquad eV_{02} = E_2 - \phi$

$eV_{02} - eV_{01} = E_2 - E_1 = 3.44 - 3.1 = 0.34\ eV$

$V_{02} - V_{01} = 0.34\ V$

4. $\frac{K_2}{K_1}=\frac{h\nu_2-\phi}{h\nu_1-\phi}$

$\frac{v_2^2}{v_1^2}=\frac{9h\nu_0-h\nu_0}{3h\nu_0-h\nu_0}=4$

$v_2=2v_1=2v$

5. $3eV_0=\frac{hc}{\lambda}-\phi$ ...(1)

$eV_0=\frac{hc}{2\lambda}-\phi$ ...(2)

$3\left[\frac{hc}{2\lambda}-\phi\right]=\frac{hc}{\lambda}-\phi$

$3\phi-\phi=\frac{3hc}{\lambda}-\frac{hc}{\lambda}$

$\Rightarrow \quad 2\phi=\frac{hc}{2\lambda} \quad \Rightarrow \quad \phi=\frac{hc}{4\lambda}$

$\Rightarrow \quad \frac{hc}{\lambda_0}=\frac{hc}{4\lambda}$

$\lambda_0 = 4\lambda$

6. $P = 1.5\ mW = 1.5 \times 10^{-3}\ W$

$\lambda = 400\ nm = 4 \times 10^{-7}\ m$

$x = 0.1\ \%$

$\frac{0.1}{100}=\frac{n_e}{n_{Ph}}$

$n_e=\frac{1}{1000}n_{Ph}$

$n_e=\frac{1}{1000}\times 5\times 10^{24}\times P\lambda$

$n_e = 5 \times 10^{21} \times 1.5 \times 10^{-3} \times 4 \times 10^{-7}$

$n_e = 3 \times 10^{12}$ photo e^-/sec

Photo current (i) = n_e e

$i = 3 \times 10^{12} \times 1.6 \times 10^{-19}$

$i = 4.8 \times 10^{-7}$

$i = 0.48\ \mu A$

7. $eX = 5h\nu - \phi$

$eX = 5h\nu - h\nu = 4h\nu$

$eX = 4h\nu$ ... (1)

Let new stopping potential be Y

$eY = 7h\nu - h\nu$

$eY = 6h\nu$ ... (2)

(2) ÷ (1)

$\frac{Y}{X}=\frac{6h\nu}{4h\nu}=\frac{3}{2}=1.5$

$Y = 1.5\ X$

8. $\phi = 2.8\ eV$

$K_{max} = 2eV$

$K_{max} = h\nu - \phi$

$2\ eV = h\nu - 2.8\ eV$

$h\nu = 4.8 \quad eV$ ... (1)

when frequency is doubled

$K'_{max} = 2h\nu - \phi$

$= (2 \times 4.8 - 2.8)\ eV$

$K'_{max} = 6.8\ eV$

9. Metal A

$h\nu = K + h\nu_0$ ... (1)

Metal B

$\phi = 2h\nu_0$

$K'_{max} = 3h\nu - 2h\nu_0$ ... (2)

Put (1) in (2)

$K'_{max} = 3(K + h\nu_0) - 2h\nu_0$

$= 3K + 3h\nu_0 - 2h\nu_0$

$K'_{max} = 3K + h\nu_0$

10. $\lambda = 4000\text{Å}$

$|V_0| = 1.4\ V$

$E=\frac{12400}{4000}=3.1\ eV$

$eV_0 \Rightarrow E - \phi$

$\phi = E - eV_0$

$\phi = 3.1 - 1.4 = 1.7\ eV$

MATTER WAVES

BEGINNER'S BOX-3

1. $\lambda=\frac{12.27}{\sqrt{V}}$

$\lambda'=\frac{12.27}{\sqrt{4V}}=\frac{12.27}{2\sqrt{V}}$

$\lambda'=\frac{\lambda}{2}$

2. $\lambda=\frac{h}{mv}=\frac{h}{\sqrt{2mE}}$

$E=\frac{3}{2}kT$

$\therefore \quad \lambda=\frac{h}{\sqrt{3mkT}}$

$\frac{\lambda_H}{\lambda_{He}}=\frac{\sqrt{M_{He}T_{He}}}{\sqrt{M_H T_H}}$

$=\sqrt{\frac{4\times(273+127)}{2\times(273+27)}}=\sqrt{\frac{8}{3}}$

3. In this process linear momentum is conserved.

$\therefore \quad |P_1| = |P_2|$

$$\lambda = \frac{h}{P} \quad \Rightarrow \quad \frac{\lambda_1}{\lambda_2} = \frac{P_2}{P_1} = 1:1$$

4. $\lambda = 5200$ Å

$$p_{Photon} = \frac{h}{\lambda} = \frac{6.6\times10^{-34}}{5200\times10^{-10}}$$

$p_{Photon} = 0.127 \times 10^{-26}$

$\because\ p_e = p_{Photon}$ (given)

$\therefore\ m_e v \Rightarrow 0.127 \times 10^{-26}$

$$v = \frac{0.127\times10^{-26}}{9.1\times10^{-31}}$$

$v = 1395.6$ m/s

5. $$\lambda = \frac{12.27}{\sqrt{V}}$$

$$V = \frac{150.6}{\lambda^2} = \frac{150.6}{(0.6)^2} = 418.33 \text{ volts}$$

6. $E = mc^2$

$$m = \frac{E}{c^2}$$

$$E = \frac{hc}{\lambda}$$

$$m = \frac{hc}{\lambda c^2} = \frac{h}{c\lambda}$$

MODERN PHYSICS-2 [Nuclear Physics and Radioactivity]

BEGINNER'S BOX-1

1. (i) $Q = [m(^1_1H) + m(^3_1H) - 2m(^2_1H)]c^2$

$= [1.007825 + 3.016049 - 2\times2.014102]c^2$

$Q = [-0.00433]c^2 = -4.031$ MeV

The –ve value of Q shows that reacⁿ is endothermic

(ii) $Q = \left[2m^{12}_6C - m\left(^{20}_{10}Ne\right) - m\left(^4_2He\right)\right]c^2$

$= [2 \times 12 - 19.992439 - 4.0015]c^2$

$= [0.006061]c^2 \quad = 5.64$ MeV

+ve value of Q shows that reacⁿ is exothermic.

2. No. of neutrons = $n_n = 35 - 17 = 18$

No. of protons = $n_P = 17$

$m_P = 1.007277$ u

$m_n = 1.008665$ u

$M_{nucleus} = 34.98$ u

$B.E. = \left[[m_P n_P + m_n n_n] - M_{nucleus}\right]c^2$

$= [17.123709 + 18.15597 - 34.98]c^2$

$= 0.299$ u $\times c^2$

$= 0.299 \times 931$ MeV

$BE = 279$ MeV

3. Number of atoms in 2gm $U^{235} = \frac{2N_A}{235}$

Energy released $= \frac{2N_A}{235} \times 200$ MeV

$$= \frac{2\times6.023\times200\times10^{23}}{235} \text{ MeV}$$

$= 10.25 \times 10^{23}$ MeV

$= 10.25 \times 10^{23} \times 10^6 \times 1.6 \times 10^{-19}$ J

$= 16.4 \times 10^{10}$ J

$$= \frac{16.4\times10^{10}}{3.6\times10^6} \text{ KWh}$$

$= 4.55 \times 10^4$ KWh

4. $P = 16$ KW $= 16 \times 10^3$ W

$$P = \frac{\text{Energy}}{\text{Time}}$$

$$= \frac{\text{Energy released per fission} \times \text{no. of fission}}{\text{Time}}$$

$$16 \times 10^3 = \frac{3.2 \times 10^{-11} \times \text{no. of fission}}{\text{Time}}$$

$$5 \times 10^{14} = \frac{\text{no. of fission}}{\text{Time}}$$

Number of nuclei per second $= 5 \times 10^{14}$

5. Energy released per fission = 200 MeV

$= 200 \times 10^6 \times 1.6 \times 10^{-19}$ J

$= 320 \times 10^{-13}$ J

Let m be mass required

Number of atoms in m mass of $U^{235} = \frac{mN_A}{235}$

Total energy released $= \frac{mN_A \times 320\times10^{-13}}{235}$ J

$$\text{Power} = \frac{\text{total energy released}}{\text{time}}$$

$$10 \times 10^6 = \left(\frac{m}{t}\right) \times \frac{6.023\times10^{23}\times320\times10^{-13}}{235}$$

$t = 1$ day $= 24 \times 60 \times 60$ sec

$$10^7 = \frac{m\times6.023\times10^{23}\times320\times10^{-13}}{86400\times235}$$

$$m = \frac{10^7\times86400\times235}{6.023\times10^{10}\times320}$$

$m = 10.5$ g

6. $\Delta m = 0.3\%$ of 1 kg

$\Delta m = \frac{3}{1000}$ kg

Energy released = $(\Delta m)c^2$

$= \frac{3}{1000} \times 9 \times 10^{16} = 27 \times 10^{13}$ J

$= 2.7 \times 10^{14}$ joule

7. $\because$ Nuclear density is independent of mass number

$\therefore$ Ratio is 1 : 1

RADIOACTIVITY

BEGINNER'S BOX-2

1. ${}_{92}U^{238} \xrightarrow{\alpha} {}_{90}X_1^{234} \xrightarrow{+\beta^0} {}_{89}X^{234}$

a = 89

b = 234

2. ${}_{a}X^{b} \xrightarrow{-\beta^0} {}_{a+1}X_1^{b} \xrightarrow{\alpha} {}_{a+1-2}Y^{b-4} \left(= {}_{a-1}Y^{b-4}\right)$

$\downarrow -\beta^0$

${}_{a-1+1}Y^{b-4} \left(= {}_{a}Y^{b-4}\right)$

a = 110

b – 4 = 215, b = 219

c = a – 1 = 109

d = b – 4 = 219 – 4 = 215

3. $N_\alpha = \frac{A_i - A_f}{4} = \frac{238-206}{4} = \frac{32}{4} = 8$

$N_\beta = Z_f - [Zi - 2\,N\alpha]$

$= 82 - [92 - 2 \times 8]$

$= 82 - [92 - 16] = 82 - 76 = 6$

4. ${}_{90}Th^{232} \xrightarrow[N_{\beta^-}]{N_\alpha} {}_{82}Pb^{200}$

$N_\alpha = \frac{A_i - A_f}{4} = \frac{232-200}{4} = \frac{32}{4} = 8$

$N_\beta = Z_f - [Z_i - 2N_\alpha]$

$= 82 - [90 - 2 \times 8] = 82 - [90 - 16]$

$= 82 - 74 = 8$

5. ${}_{72}A^{180} \xrightarrow{\alpha} {}_{70}A_1^{176} \xrightarrow{B^-} {}_{71}A_2^{176}$

$\downarrow \alpha$

${}_{69}A_4^{172} \xleftarrow{\gamma} {}_{69}A_3^{172}$

6. (i) ${}^{226}_{88}Ra \xrightarrow{-\alpha} {}^{222}_{86}Rn + {}^{4}_{2}He$

(ii) ${}^{242}_{94}Pu \xrightarrow{-\alpha} {}^{238}_{92}U + {}^{4}_{2}He$

(iii) ${}^{32}_{15}P \xrightarrow{-\beta^-} {}^{32}_{16}S + {}^{0}_{-1}\beta + \bar{\upsilon}$

(iv) ${}^{210}_{83}Bi \xrightarrow{-\beta^-} {}^{210}_{84}Po + {}^{0}_{-1}\beta + \bar{\upsilon}$

(v) ${}^{11}_{6}C \xrightarrow{-\beta^+} {}^{11}_{5}B + {}^{0}_{+1}\beta + \upsilon$

(vi) ${}^{97}_{43}Tc \xrightarrow{-\beta^+} {}^{97}_{42}Mo + {}^{0}_{+1}\beta + \upsilon$

(vii) ${}^{120}_{54}Xe + {}_{-1}e^{0} \xrightarrow{\text{electron capture}} {}^{120}_{53}I + \text{X-Ray} + \upsilon$

MODERN PHYSICS-3 [Atomic Structure]

BEGINNER'S BOX-1

1. $\lambda = 975 \times 10^{-10}$ m

$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{n^2}\right]$

$\frac{1}{975 \times 10^{-10}} = 1.1 \times 10^7 \left[\frac{1}{1^2} - \frac{1}{n^2}\right]$

n = 4

λ is minimum for transition from $4 \to 1$

this is equivalent to wavelength of incident photon

$\lambda_{min} = 975$Å

λ is maximum for transition from $4 \to 3$

Energy emitted for this transition is

$E_4 - E_3 = -(.85) - (-1.51)$

$= .66$ eV

$\lambda_{max} = \frac{12400}{.66} = 18787.87$Å

2. $\lambda = 975 \times 10^{-10}$ m

$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{n^2}\right]$

$\frac{1}{975 \times 10^{-10}} = (1.1 \times 10^7)\left[1 - \frac{1}{n^2}\right]$

n = 4

But number of absorption lines = 1(in the transition from 1 to 4)

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3. $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$\frac{1}{\lambda_{Li}} = R \times 3^2 \times \left(1 - \frac{1}{2^2}\right)$

$\frac{1}{\lambda_H} = R \times 1\left(1 - \frac{1}{2^2}\right)$

$\frac{\frac{1}{\lambda_{Li}}}{\frac{1}{\lambda_H}} = \frac{9}{1}$

$\frac{\lambda_H}{\lambda_{Li}} = \frac{9}{1}$ $\qquad$ $\frac{\lambda_{Li}}{\lambda_H} = \frac{1}{9}$

4. $r \propto \frac{n^2}{Z}$

for n = 1 $\quad$ r = r_1

n = 2 $\quad$ r = r_2

$\frac{r_2}{r_1} = \frac{n_2^2}{n_1^2} = \frac{4}{1} \Rightarrow \frac{A_2}{A_1} = \frac{\pi r_2^2}{\pi r_1^2} \Rightarrow \frac{16}{1}$

5. I.E. = 13.6 eV

I.E. = $-E_1$

E_1 = –13.6 eV

Excitation potential of 3rd orbit is $E_4 - E_3$

$= -\frac{13.6}{4^2} - \left(\frac{-13.6}{3^2}\right)$

= –(.85) – (–1.51) $\Rightarrow$ 0.66 V

6. $\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = 1.1 \times 10^7\left(1 - \frac{1}{2^2}\right)$

$\Rightarrow 0.825 \times 10^7$

$\lambda = \frac{1}{.825 \times 10^7} = 1.21 \times 10^{-7} = 1210\text{Å} = 121 \text{ nm}$

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Wave Optics

SOLUTIONS

BEGINNER'S BOX-1

1. Resultant intensity

$$I_R = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$$

putting the values

$$I_R = I + 4I + 2\sqrt{I_1}\sqrt{4I}\cos\frac{\pi}{2} = 5I$$

2. $$\frac{I_{max}}{I_{min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2 = \left(\frac{\sqrt{\frac{I_1}{I_2}}+1}{\sqrt{\frac{I_1}{I_2}}-1}\right)^2$$

$$= \left(\frac{\sqrt{81}+1}{\sqrt{81}-1}\right)^2 = \left(\frac{9+1}{9-1}\right)^2 = \frac{100}{64} = \frac{25}{16}$$

3. $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\theta$

(a) $(I_R)_1 = I + 4I + 2\sqrt{I\times 4I}\cos 45° = 7.8I$

(b) $(I_R)_2 = I + 4I + 2\sqrt{I\times 4I}\cos 180°$

$= I + 4I - 4I = I$

(c) $(I_R)_3 = I + 4I + 2\sqrt{I\times 4I}\cos 4\pi = 9I$

BEGINNER'S BOX-2

1. Let n^{th} bright fringe of one coincide with m^{th} bright fringe of the other, then :

$$\frac{n\lambda D}{d} = \frac{m\lambda' D}{d} \Rightarrow \frac{n}{m} = \frac{\lambda'}{\lambda} = \frac{9000}{7500} = \frac{6}{5}$$

Hence for first position at which the overlap occurs is for

n = 6 and m = 5 and position Y is

$$Y = \frac{n\lambda D}{d} = \frac{5\times 9000\times 10^{-10}\times 2}{2\times 10^{-3}} = 4.5 \text{ mm}$$

2. Angular width, $\theta = \frac{\beta}{D}$, But $\beta = \frac{\lambda D}{d}$

$$\therefore \quad \theta = \frac{\lambda}{d} \Rightarrow d = \frac{6000}{(0\cdot 1)}\times 10^{-10} = 6\mu m$$

3. Distance of fifth bright fringe

$$x_5 = \frac{5\lambda D}{d} = \frac{5\times 80\times 6\times 10^{-5}}{0\cdot 02} = 1\cdot 2 \text{ cm}$$

4. Position of 8th Bright fringe $\frac{n\lambda D}{d} = \frac{8\lambda D}{d}$,

Position of 3rd dark fringe $= \frac{(2n-1)\lambda D}{2d}$

$$= \frac{5\lambda D}{2d} = \frac{25\lambda D}{d}$$

Separation between 8th and 3rd fringes

$$= \frac{8\lambda D}{d} - \frac{2.5\lambda D}{d} = \frac{5.5\lambda D}{d} = \frac{5.5\times 480\times 10^{-9}\times 2}{3\times 10^{-3}}$$

$= 1.76 \times 10^{-3}$ m

5. $\beta = \frac{\lambda D}{d} \Rightarrow \Delta\beta = \frac{\lambda\Delta D}{d}$

or $\lambda = \frac{(\Delta\beta)d}{\Delta D} = \frac{3\times 10^{-5}\times 10^{-3}}{5\times 10^{-2}} = 6 \times 10^{-7}$ m

6. fringe width $\beta_{air} = \frac{\lambda D}{d} = \frac{1\times 5896\times 10^{-10}}{2\times 10^{-3}}$

$= 0.2948$ mm

Now the system is immersed in water, then :

$$\frac{\beta_w}{\beta_{air}} = \frac{\lambda_w}{\lambda_{air}} = \frac{\mu_{air}}{\mu_w} = \frac{1}{\mu_w}$$

$$\Rightarrow \beta_w = \frac{1}{4/3}\times 0.2948 = 0.2211 \text{ mm}$$

7. For dark $x_n = \frac{3\lambda D}{2d}$ (Second dark band)

$$\Rightarrow \lambda = \frac{2x_n d}{3D} = \frac{2\times 3\times 10^{-3}\times 3\times 10^{-4}}{3\times 0.9}$$

$$\Rightarrow \lambda = \frac{2}{3}\times 10^{-6} \text{ m}$$

For 4th Bright fringe

$$x = \frac{4\lambda D}{d} = \frac{4\times 2\times 10^{-6}\times 0.9}{3\times 3\times 10^{-4}} = 8 \times 10^{-3} \text{ m}$$

8. fringe width $\beta = \frac{\lambda D}{d}$ Here,

$$\beta_1 = \frac{\lambda_1 D_1}{d_1} \text{ and } \beta_2 = \frac{\lambda_2 D_2}{d_2}$$

As $\beta_1 = \beta_2 \Rightarrow \frac{\lambda_1 D_1}{d_1} = \frac{\lambda_2 D_2}{d_2}$

$$\Rightarrow \frac{D_1}{D_2} = \frac{\lambda_2 d_1}{\lambda_1 d_2} = \frac{600}{400}\times\frac{1}{1/2} = \frac{6}{2} = \frac{3}{1}$$

9. Fringe width $\beta = \frac{\lambda D}{d} = \frac{c\ D}{\nu\ d}$

$\therefore\ d = \frac{c\ D}{\nu\ \beta} = \frac{3\times10^{8}\times1.5}{6\times10^{14}\times75\times10^{-5}} = 10^{-3}$ m.

10. $\beta_{\ell} = \frac{\beta}{\mu} = \frac{2}{1.33} = 1.5$mm

BEGINNER'S BOX-3

1. On placing a thin sheet of mica additional path difference in interferring beams.

$\Delta x = (\mu - 1)t = n\lambda$

Now according to question

$(\mu - 1)\ 12 \times 10^{-7} = 1 \times 6 \times 10^{-7}$

$\Rightarrow \quad \mu = \frac{6\times10^{-7}}{12\times10^{-7}} + 1 = \frac{1}{2} + 1 = \frac{3}{2} = 1{\cdot}5$

2. $x = \frac{(\mu-1)t\,D}{d} = \frac{(\mu-1)t\,\beta}{\lambda}$

$\because\ \beta = \frac{\lambda D}{d} \Rightarrow \frac{\beta}{\lambda} = \frac{D}{d}$ and $x = 5\beta$

$\therefore\ 5\beta = \frac{(\mu-1)t\,\beta}{\lambda} \Rightarrow t = \frac{(\mu-1)t\,\beta}{\lambda}$

$\Rightarrow \frac{5\times6000\text{Å}}{(1.5-1)} \Rightarrow 6 \times 10^{-6}$ m

3. For bright fringe in transmitted light

$2\mu t \cos\theta = n\lambda$ & for minimum thickness $n = 1$

Therefore $t_{min} = \frac{\lambda}{2\mu\cos30°} = \frac{6\times10^{-5}}{2\times\frac{4}{3}\times\frac{\sqrt{3}}{2}}$ cm

$= 2.59 \times 10^{-5}$ cm

4. For dark fringe in reflected light

$2\mu t \cos r = n\lambda \Rightarrow t_{min} = \frac{\lambda}{2\mu\cos r}$

so $t_{min} = \frac{5500}{2\times\frac{4}{3}\times\cos60°} = \frac{5500}{2\times\frac{4}{3}\times\frac{1}{2}}$

$= 4125$ Å

BEGINNER'S BOX-4

1. Width of central maxima

$W_x = \frac{2\lambda D}{a} = \frac{2\times2{\cdot}1\times5\times10^{-7}}{0{\cdot}15\times10^{-2}}$

$\Rightarrow \quad W_x = 1{\cdot}4$ mm

2. $\theta = \frac{2\lambda}{a} = \frac{2\times600\times10^{-9}}{3\times10^{-3}}$

$= 0.4 \times 10^{-3}$ radian $= 0.023°$

3. $W_x = \frac{2\lambda D}{a} = \frac{2\times6500\times10^{-10}\times1.8}{0.5\times10^{-3}}$

$= 468 \times 10^{-5}$ m $= 4.68$ mm

4. First minimum and first maximum coincide

$\theta_1 = \theta_2$ or $\sin\theta_1 = \sin\theta_2$

$\therefore \quad \frac{\lambda_1}{d} = \frac{(2n+1)\lambda_2}{2d}$

$\Rightarrow \quad \lambda_2 = \frac{2}{3}\lambda_1 = \frac{2}{3} \times 660$ nm $= 440$ nm.

BEGINNER'S BOX-5

1. $\mu = 1.33, \quad \theta_P = ?$

$\mu = \tan\theta_P$

$1.33 = \tan\theta_P \Rightarrow \quad \theta_P = \tan^{-1}\left(\frac{4}{3}\right)$

$\theta_P = 53°4'$

2. $\mu = \tan\theta_P$

$\mu = \tan60° \Rightarrow \quad \mu = \sqrt{3} = 1.732$

3. $\theta_P = 60°$

$\mu = \tan60° \Rightarrow \mu = \sqrt{3}$

$\mu = \frac{\sin i}{\sin r} \Rightarrow \sqrt{3} = \frac{\sin60°}{\sin r}$

$\sqrt{3} = \frac{\sqrt{3}}{2\sin r} \Rightarrow \sin r = \frac{1}{2}$

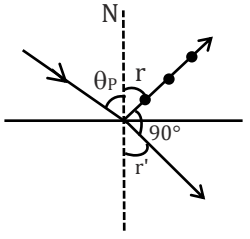
$\boxed{r = 30°}$

(a) $r = 30°$ (angle of refrection), (b) $\mu = \sqrt{3}$

4. $\mu = 1.732$ (given)

$\mu = \tan\theta_P \Rightarrow 1.732 = \tan\theta_P$

$\theta_P = 60°$



glass ($\mu = 1.57$)

Here, $r = \theta_P$

$\theta_P + 90° + r' = 180°$

$60° + 90° + r' = 180°$

$r' = 30°$

5. $\mu = \frac{1}{\sin\theta_C} = \frac{1}{\sin40°}$ (θ_c = critical angle)

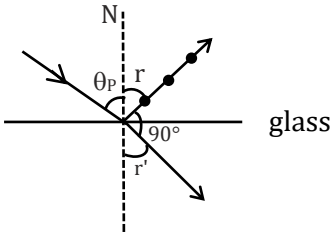
$\mu = \frac{1}{2/3} = \frac{3}{2}$

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Now, $\mu = \tan\theta_p$ (θ_p = polarising angle)

$$\frac{3}{2} = \tan\theta_p \Rightarrow \theta_p = \tan^{-1}(1.5)$$

$\theta_p = 57.3° = r$



r' = 180° – (90° + r)

r' = 180° – (90° + 57.3°)

r' = 32.7°

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Ray Optics and Optical Instrument

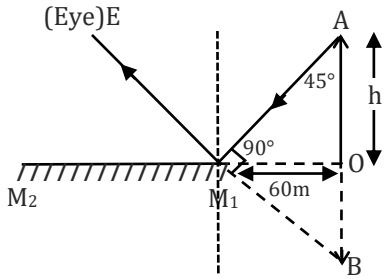
SOLUTIONS

BEGINNER'S BOX-1

1. Minimum length of the mirror $= \frac{H}{2}$

$= \frac{1.8}{2} = 0.9$ m

3.



From figure $\angle AM_1E = 90°$

$\therefore \quad \angle AM_1B = 90°$

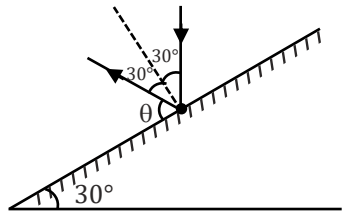
$\angle A = \angle B = 45°$

In ΔAM_1O

$\tan 45° = \frac{60}{OA} = \frac{60}{h}$

$\therefore \quad h = 60m$

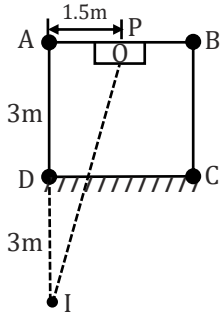
4.



Here $90° - \theta = 30°$

$\Rightarrow \quad \theta = 60°$

5.



Distance of image from camera

$= \sqrt{(6)^2 + (1.5)^2} = 6.18$ m

BEGINNER'S BOX-2

2. In given conditions $m = -\frac{v}{u} = 2 \Rightarrow v = -2u$

from mirror equation $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

given $f = -0.2$ m

$\Rightarrow \quad \frac{1}{-2u} + \frac{1}{u} = \frac{1}{-0.2} \Rightarrow u = -0.1m$

so mirror should be placed at the distance of 0.1 m.

3. From mirror equation $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \quad \frac{1}{v} + \frac{1}{-f/2} = \frac{1}{f} \Rightarrow v = \frac{f}{3}$

and $m = -\frac{v}{u} = -\frac{f/3}{-f/2} = \frac{2}{3}$ so image will be virtual, erect and smaller than object

4. From mirror equation $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \quad \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} - \frac{1}{-600} = \frac{31}{600}$

$\Rightarrow \quad v = \frac{600}{31} = 19.35$ cm behind the mirror.

5. $\because \quad m = -\frac{v}{u} = 3 \Rightarrow v = -3u$

Now $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

where $f = \frac{-R}{2} = \frac{-36}{2} = -18\,cm$

$\Rightarrow \quad \frac{1}{-3u} + \frac{1}{v} = \frac{1}{-18} \quad \Rightarrow \quad \frac{-1+3}{3u} = \frac{1}{-18}$

$\Rightarrow \quad u = -12$ cm

6. $f = -30$ cm & $m = -3$

$m = \frac{f}{f - u} \quad \therefore \quad -3 = \frac{-30}{-30 - u}$

$\Rightarrow \quad u = -40$ cm

$\Rightarrow \quad u = 40$ cm in front of mirror

7. R = –36 cm, f = –18 cm

$m = \frac{f}{f-u} = \frac{-18}{-18-(-27)} = \frac{-18}{9} = -2$ (inverted)

$m = \frac{-v}{u} \Rightarrow -2 = \frac{-v}{-27}$

$\Rightarrow$ v=–54 cm (Real image)

$m = \frac{h_i}{h_0} \Rightarrow h_i = m \times h_0$

$\Rightarrow h_i = -2 \times 2.5$

$= -5$cm

BEGINNER'S BOX-3

2. given $i = \frac{r}{2}$ so from snell's law $\mu \sin i = \sin r$

$\mu \sin\left(\frac{r}{2}\right) = 2\sin\left(\frac{r}{2}\right)\cos\left(\frac{r}{2}\right)$

$\cos\left(\frac{r}{2}\right) = \frac{\mu}{2}$

$\Rightarrow r = 2\cos^{-1}\left(\frac{\mu}{2}\right)$

BEGINNER'S BOX-4

1. Refractive index $\mu = \frac{v_{air}}{v_{medium}} = \frac{\frac{x}{t_1}}{\frac{10x}{t_2}} = \frac{t_2}{10t_1}$

also $\sin\theta_c = \frac{1}{\mu} = \frac{10\ t_1}{t_2} \Rightarrow \theta_c = \sin^{-1}\left(\frac{10t_1}{t_2}\right)$

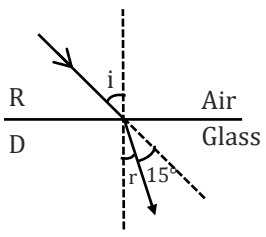
2. $\delta = i - r$, $15° = 45° - r$, $r = 30°$

$\mu_{air} \sin i = \mu_g \times \sin r$

$1 \times \sin 45° = \mu_g \times \sin 30°$

$\frac{1}{\sqrt{2}} = \mu_g \times \frac{1}{2}$

$\mu_g = \frac{2}{\sqrt{2}} = \sqrt{2}$



$\sin\theta_c = \frac{1}{\mu_g} = \frac{1}{\sqrt{2}}$

$\theta_C = 45°$

3. Angle of incidence (i) < critical angle (θ_c)

$\sin\theta_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}} \Rightarrow \theta_C = 45°$

For ray to emerge out $i < \theta_C$ OR $i < 45°$

BEGINNER'S BOX-5

1. $\because \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$

$\therefore \frac{1}{v} - \frac{1.5}{(-5)} = \frac{(1-1{\cdot}5)}{-20} \Rightarrow$ v = + 3.63cm

2. Let v be the apparent depth of bubble then by using

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$\therefore \frac{1}{v} - \frac{1.5}{-4} = \frac{1-1.5}{-10}$

$\Rightarrow v = -\frac{40}{13} = -3.07$ cm

BEGINNER'S BOX-6

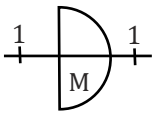
1. $m = \frac{f}{f+u} = \frac{10}{10-30} = \frac{10}{-20} = \frac{-1}{2}$ (Inverted)

$m = \frac{v}{u} \Rightarrow \frac{-1}{2} = \frac{v}{-30}$

v = +15 cm (Real)

2. Lens maker formula $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\Rightarrow \frac{1}{30} = (1.50-1)\left(\frac{1}{\infty} - \frac{1}{-R}\right)$



$\Rightarrow \frac{1}{30} = \frac{0.50}{+R}$

$\Rightarrow$ R = + 15 cm

3. $\mu = 1.5$, $P_m = -1 = \frac{\mu_m}{f_m}$, $f_{air} = 0.2$ m, $\mu_m = ?$

$\frac{1}{f_m} = \left(\frac{\mu}{\mu_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\frac{1}{f_{air}} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Divide both equation

$\frac{f_{air}}{f_m} = \frac{\left(\frac{\mu}{\mu_m} - 1\right)}{(\mu - 1)} \Rightarrow \frac{0.2}{-\mu_m} = \frac{\left(\frac{\mu}{\mu_m} - 1\right)}{(\mu - 1)}$

$0.2\mu - 0.2 = -\mu + \mu_m$

$1.2\ \mu = 0.2 + \mu_m$

$1.2 \times 1.5 = 0.2 + \mu_m$

$\mu_m = 1.6$

4. Power of combination

$P = \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{100}{20} + \frac{100}{40} = 5 + 2.5 = 7.5\ D$

5. $u = -20$ cm, $v = +40$ cm ;

$P = \frac{1}{f} = \frac{100}{v} - \frac{100}{u} = \frac{100}{40} - \frac{100}{-20} = 7.5\ D$

6. Focal length changes due to change in refractive index hence there will be 2 images.

7. Image is real so $m = \frac{v}{u} = -\frac{1}{n}$

$\Rightarrow v = -\frac{u}{n}$

from lens formula

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-\frac{u}{n}} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{-(n+1)}{u} = \frac{1}{f} \Rightarrow u = -f(1+n)$

BEGINNER'S BOX-7

1. Using displacement method

$f = \frac{D^2 - d^2}{4D} \Rightarrow 16 = \frac{(70)^2 - d^2}{4 \times 70}$

$d^2 = 70(70-64)$

$d^2 = 70 \times 6 = 420$

$d = 20.5$ cm

2. $F = 90$cm, $\omega_1 = 1.5\omega_2$ $\therefore \frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2} = -\frac{3}{2}$

$\Rightarrow f_1 = -\frac{3}{2} f_2$

By using $\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{F}$

we get $f_1 = -45$ cm and $f_2 = 30$ cm

BEGINNER'S BOX-8

1. Deviation $\delta_{min} = 2i - A = 2 \times 45° - 60°$

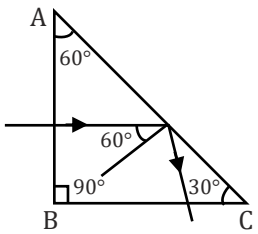
$= 90° - 60° = 30°$

2. For no emergence,

$r_2 > \theta_C$

$\sin 60° > \sin \theta_C$

$\Rightarrow \frac{1}{\mu} < \frac{\sqrt{3}}{2} \Rightarrow \mu > \frac{2}{\sqrt{3}}$



3. Refractive index of material of prism

$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{60° + 60°}{2}\right)}{\sin\left(\frac{60°}{2}\right)}$

$\mu = \frac{\sin 60°}{\sin 30°} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$

4. In condition of minimum deviation

$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin(A/2)}$

$\because i = 2r \therefore \delta_m = 2i - 2r = 2(2r) - 2r = 2r = A$

therefore $\mu = \frac{\sin A}{\sin A/2} = 2\cos\frac{A}{2}$

$\Rightarrow \cos\frac{A}{2} = \frac{\mu}{2} = \frac{\sqrt{2}}{2}$

$\Rightarrow \frac{A}{2} = 45°$

$\Rightarrow A = 90°$

5. $\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\ (A/2)} = \frac{\sin\left(\frac{60°+30°}{2}\right)}{\sin\left(\frac{60}{2}\right)}$

$\mu = \frac{\sin 45°}{\sin 30°} = \frac{\frac{1}{\sqrt{2}}}{1/2} = \sqrt{2}$

6. $i_1 = 15°$, $A = 30°$, $\delta = 55°$, $i_2 = ?$

$i_1 + i_2 = A + \delta$; $15 + i_2 = 30 + 55$

$\Rightarrow i_2 = 15 + 55 = 70°$

BEGINNER'S BOX-9

1. $M_{net} = m_o \times m_e$ .....(i)

$m_e = 1 + \frac{D}{f_e}$

$m_e = 1 + \frac{25}{5} = 6$

According to equation (i)

$30 = m_0 \times 6 \Rightarrow m_0 = 5$

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2. $v_e = -25\text{cm}, f_e = \frac{100}{P_e} = \frac{100}{20} = 5\text{ cm},$

$f_0 = \frac{100}{P_0} = \frac{100}{0.5} = 200\text{ cm}$

from $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow \frac{1}{-25} - \frac{1}{u_e} = \frac{1}{5}$

$\Rightarrow \quad u_e = -\frac{25}{6}$

$L = f_0 + |u_e| = 200 + \frac{25}{6} = 204.17\text{ cm}$

3. For Galilean telescope in normal adjustment

$M = \frac{f_0}{f_e} = \frac{30}{5} = 6$

$L = f_0 - f_e = 30 - 5 = 25\text{ cm}.$

4. M.P. = $\frac{f_o}{f_e} = \frac{\beta}{\alpha}$ [Final image at ∞]

$\beta = \alpha \times \frac{f_o}{f_e} = 2° \times \frac{60}{5} = 24°$

5. (i) When the final image is at infinity then

$\text{M.P.} = \frac{-f_o}{f_e} = \frac{-60}{5} = -12$

and length of the telescope is

$L = |f_0| + |f_e|$

$L = f_o + f_e = 60 + 5 = 65\text{ cm}$

(ii) For least distance of distinct vision, the magnifying power is

$\text{M.P.} = \frac{-f_o}{f_e}\left(1 + \frac{f_e}{D}\right) = \frac{-60}{5}\left(1 + \frac{5}{25}\right)$

$= \frac{-12 \times 6}{5} = -14.4$

Now $\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e} \Rightarrow \frac{1}{5} = \frac{-1}{25} - \frac{1}{u_e}$

$\frac{-1}{u_e} = \frac{1}{25} + \frac{1}{5}$

$u_e = -4.17\text{ cm}$

$|u_e| = 4.17\text{ cm}$

The length of the tube of a telescope

$L = |f_o| + |u_e|$

$L = 60 + 4.17 = 64.17\text{ cm}$

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Alternating Current

SOLUTIONS

Beginner's Box - 1

1. There are few reasons for it :
- A.C. attracts while D.C. repels.
- A.C. gives a huge and sudden shock.
- Frequency (50 to 60 Hz) shocks are injurious to human body.

2. For AC, $I = I_0 \sin\omega t$, the instantaneous value of heat produced (per second) in a resistance R is, $H = I^2R = I_0^2\sin^2\omega t \times R$ the average value of heat produced during a cycle is :

$$H_{av} = \frac{\int_0^T H\,dt}{\int_0^T dt} = \frac{\int_0^T (I_0^2\sin^2\omega t \times R)dt}{\int_0^T dt} = \frac{1}{2}I_0^2 R$$

$$\left[\because \int_0^T I_0^2\sin^2\omega t\,dt = \frac{1}{2}I_0^2 T\right]$$

$$\Rightarrow H_{av} = \left(\frac{I_0}{\sqrt{2}}\right)^2 R = I_{rms}^2 R \text{ .....(i)}$$

However, in case of DC, $H_{DC} = I^2 R$...(ii)

$\because I = I_{rms}$ so from equation (i) and (ii) $H_{DC} = H_{av}$

AC produces same heating effects as DC of value $I = I_{rms}$. This is also why AC instruments which are based on heating effect of current give rms value.

3. The average value of a.c. for a cycle is zero. So a d.c. ammeter will always read zero in a.c. circuit.

4. $I = I_0 \sin\omega t$ $\qquad \frac{I_0}{\sqrt{2}} = I_0 \sin\omega t$

$\omega t = \frac{\pi}{4}$ $\qquad t = \frac{\pi \times T}{4 \times 2\pi} = \frac{T}{8} = \frac{1}{50 \times 8}$

$= 2.5 \times 10^{-3}$ s

5. (a) $i = 3.5 \sin(628t + 30°)$

$V = 28 \sin(628t - 30°)$

$\omega = \frac{2\pi}{T}$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{628} = 0.01\,\text{sec}$

(b) phase difference $\phi = (\phi_1 - \phi_2)$

$= 30° - (-30°) = 60°$

6. Average value for 0 to $\frac{T}{2}$ time $= \frac{2V_0}{\pi}$

$= \frac{2 \times 314}{3.14} = 200$ V

Beginner's Box - 2

1. $V_{rms} = \frac{60}{\sqrt{2}} \Rightarrow R = 20\Omega$

$I_{rms} = \frac{60}{\sqrt{2} \times 20} = \frac{3}{\sqrt{2}} = 2.1A$

2. $E = 100 \sin(1000\,t)$, $X_L = 10 \times 10^{-6} \times 1000 = 10^{-2}\Omega$

$I = \frac{100}{10^{-2}} \sin\left(1000\,t - \frac{\pi}{2}\right)$

3. $E = 200\sqrt{2} \sin 100t$

$X_L = 2 \times 100 = 200\ \Omega$

$I_{rms} = \frac{200\sqrt{2}}{\sqrt{2} \times 200} = 1$ Amp

4. $X_C = 212\ \Omega$

$I_{rms} = 1.03$ A

$I_0 = 1.46$ A

If the frequency is doubled the capacitive reactances is halved the consequently the current is doubled.

5. $Z = X_C = \frac{1}{60 \times 2\pi \times 60 \times 10^{-6}}$

$I_{rms} = \frac{V_{rms}}{Z} = \frac{110}{10^4} \times 2\pi \times 36 = 2.49$ A

6. (a) Opposition to current is constant with frequency then circuit will be resistive Circuit element – resistor.

(b) In graph b oppose to current $\propto$ f

$Z \propto f$

So circuit element inductor ($X_L = 2\pi f$)

7. $\phi = \frac{\pi}{3}$; $\tan\phi = \frac{X_L}{R}$

$\frac{X_L}{R} = \sqrt{3}$, $X_L = \sqrt{3}R$ ...(i)

$Z = \sqrt{R^2 + X_L^2}$; $10 = \sqrt{R^2 + X_L^2}$

$3R^2 + R^2 = 100$;

$4R^2 = 100$

$R = 5\Omega$ and $X_L = 5\sqrt{3}\ \Omega$,

8. $\phi = \frac{\pi}{4}$ $\tan\phi = \frac{X_L}{R}$

$1 = \frac{X_L}{R}$ $X_L = R = 314\Omega$

$\omega L = 314\ \Omega$ $314 \times L = 314$

$L = 1H$

9. Resistance of coil $R = \frac{12}{4} = 3\Omega$

When connected to A.C.

Impedence $Z = \frac{12}{2.4} = 5\Omega$

$Z^2 = R^2 + X_L^2$

$25 = 9 + X_L^2$

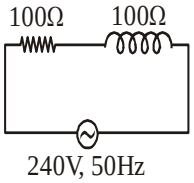
$X_L = 4\Omega \Rightarrow \omega L = 4$

$L = \frac{4}{\omega} = \frac{4}{50} = 0.08H$

10. $Z = \sqrt{R^2 + X_L^{\ 2}}$

$Z = \sqrt{100 + (800\times10^{-3}\times300)^2}$; $Z = 240.2\Omega$

11.



$Z = \sqrt{100^2 + 100^2} = 100\sqrt{2}\Omega$

$V_0 = \sqrt{2}V_{RMS} = \sqrt{2}(240)V$

(a) Maximum current

$I_0 = \frac{V_0}{Z} = \frac{\sqrt{2}(240)}{100\sqrt{2}} = 2.4amp$

(b) $\tan\phi = \frac{X_L}{R} = 1$ $\phi = \omega t$

$\phi = 45° \Rightarrow \frac{\pi}{4} = \frac{2\pi}{T}.t$

$\Rightarrow t = \frac{T}{8} = 2.5\times10^{-3}$ sec

12. Resistance of inductor $R = 100\Omega$

phase difference at frequency of 1000 Hz = 45°

$\tan\phi = \frac{X_L}{R} = 1$

$X_L = R$

$X_L = 100 \Rightarrow 2\pi fL = 100$

$L = \frac{100}{2\pi\times1000} = 15.9$ mH

13. For circuit A $\tan\phi = \frac{X_C}{R}$; $\tan\phi = \frac{1}{\omega CR}$

$= \frac{1}{200\times500\times10^{-6}\times10} = \frac{1}{1} = 1$

$\phi = \frac{\pi}{4}$

for circuit B; $\phi = \frac{\pi}{2}$ for circuit C; $\phi = \frac{\pi}{2}$

for circuit D; $\phi = \frac{\pi}{2}$ (Pure L – C circuit)

for circuit E; $\tan\phi = \frac{X_L}{R} = \frac{200\times5}{1000} = 1$

$\phi = \frac{\pi}{4}$

Beginner's Box - 3

1. $I = \frac{V}{Z} = \frac{50}{\sqrt{2}\times10\sqrt{2}} = 2.5A$

$X_L - X_C = 10\Omega$

Potential drop at combination of L & C

$= i(X_L - X_C) = 2.5 \times 10 = 25$ V

2. The three current equations are, $V = i_R R$,

$V = L\frac{di_L}{dt}$ and $\frac{dV}{dt} = \frac{1}{C}i_C$ $\frac{dV}{dt} = \frac{1}{C}i_C$

so $i_R = \frac{V_0}{R}\sin\omega t$,

$i_L = -\frac{V_0}{\omega L}\cos\omega t$ and $i_C = V_0\omega C\cos\omega t$

3. (a) $\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5\times80\times10^{-6}}} = 50$ rad/s

(b) $I_{max} = \frac{V}{R} = \frac{230}{40} = 5.75$ Amp.,

$I_0 = \sqrt{2}\ (5.75) = 8.13$ Amp.

4. $\omega_0 = \frac{1}{\sqrt{LC}}$

5. $\cos\phi = \frac{R}{Z}$

$Z = \sqrt{2}\ R$ at half power frequencies

$\cos\phi = \frac{R}{\sqrt{2}R}$; $\phi = \frac{\pi}{4}$

6. (a) Resonance frequency

$f_R = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.12\times480\times10^{-9}}}$

$= 6.63 \times 10^2$ Hz

(b) $Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{23}\sqrt{\frac{0.12}{480\times10^{-9}}} = 21.7$

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Beginner's Box - 4

1. $P = VI\cos\phi$

$\cos\phi = \frac{P}{VI} = \frac{600}{5\times160} = \frac{3}{4}$

2. (a) $H = i^2Rt$

$i = \frac{V}{Z} = \frac{25}{\sqrt{2}\times5} = \frac{5}{\sqrt{2}}$ Amp

$H = \left(\frac{5}{\sqrt{2}}\right)^2 \times 4 \times 80 = 4000$ J

(b) Wattless current = $I\sin\phi = \frac{5}{\sqrt{2}}\times\frac{3}{5} = 2.12$ Amp

3. $\phi = \frac{\pi}{3}$

$\tan\phi = \frac{X}{R} \Rightarrow \frac{X}{R} = \sqrt{3}$

$X = \sqrt{3}R$

Impedence Z of this circuit = $\frac{V}{I} = 1k\Omega$

$Z = \sqrt{X^2 + R^2}$

$10^6 = R^2 + X^2 \Rightarrow 10^6 = R^2 + 3R^2$

$R = \frac{1}{2}\times10^3 = 500\,\Omega$

So = $x = \sqrt{3}R$; $X = 500\sqrt{3}\ \Omega$

(b) $P = VI\cos\phi$

$= \frac{100}{\sqrt{2}}\times\frac{100}{\sqrt{2}}\times\frac{1}{2}\times10^{-3} = 2.5$ watt

4. (a) $Z = \frac{V}{I} = \frac{110}{5} = 22\Omega$

(b) $\phi = \frac{\pi}{3}$ (Nature inductive)

Power factor = $\cos\phi = \cos\frac{\pi}{3} = \frac{1}{2}$

5. (a) $\phi = 60°$

$\cos\phi = \frac{R}{Z}$

$\frac{1}{2} = \frac{R}{Z} \Rightarrow Z = 2R = 200\Omega$

$I = \frac{V}{Z} = \left(\frac{200}{\sqrt{2}\times200}\right) = \frac{1}{\sqrt{2}}$ Amp

(b) $P = VI\cos\phi$

$= \frac{200}{\sqrt{2}}\times\frac{1}{\sqrt{2}}\times\frac{1}{2} = 50$ Watt

6. Wattless current = $I\sin\phi = \frac{100}{5}\times\frac{4}{5} = 16$Amp

7. (a) $I_{rms} = \frac{220}{100} = 2.2$Amp

(b) Net power consumed $P = i^2R$

$= 2.2 \times 2.2 \times 100$

$= 484$ Watt

8. $V^2 = V_L^2 + V_R^2$

$V_L^2 = V^2 - V_R^2$

$V_L^2 = 130^2 - 50^2 = (130 + 50)(130 - 50)$

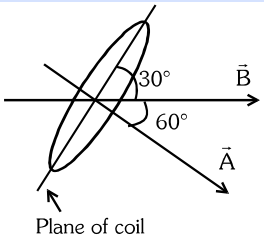
$= 180 \times 80$; $V_L = 120$ V

Electromagnetic Induction

SOLUTIONS

Beginner's Box - 1

1.



Magnetic flux $\phi = NBA\cos\theta$

$= 100 \times 0.2 \times 5 \times 10^{-4} \times \cos 60°$

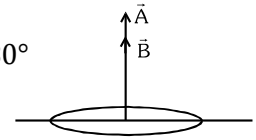
$= 5 \times 10^{-3}$ weber.

2. Initially coil is placed in transverse field

$\phi_i = NBA\ (\theta=0°)$

After rotation through 180°

$\phi_f = -NBA\ (\theta=180°)$

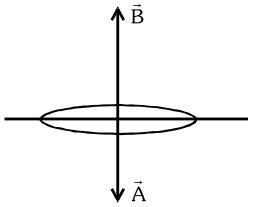


Rate of change of magnetic flux through the coil

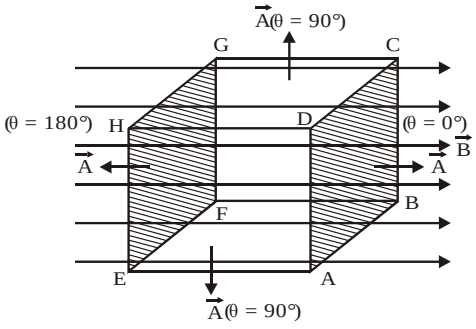
$$\frac{\Delta\phi}{\Delta t} = \frac{\phi_f - \phi_i}{\Delta t}$$

$$= \frac{-NBA - NBA}{2}$$

$= -$ NBA Wb/sec.



3.



Face	Flux
ABCD	$+Ba^2$ (out flux)
EFGH	$-Ba^2$ (in flux)
Remaining Faces	Zero

$\phi_{in} = \phi_{out}$

$\boxed{\phi_{net} = 0}$

4. Magnetic flux :–

$\phi = BA\cos\theta$

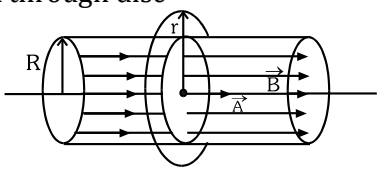
$= 0.2 \times 0.1 \times \cos 0° = 0.02$ Wb

5. Direction of area vector of loop will be perpendicular to x – y plane.

$\vec{A} = 0.1\hat{k},\ \vec{B} = (4\hat{i} - \hat{k})$

$\phi = \vec{B}.\vec{A} = (4\hat{i} - \hat{k}).(0.1\hat{k}) = -0.1$wb

6. Flux through disc



$\phi = BA\cos\theta$, where $B = \mu_0 nI$, $\theta = 0°$

$= BA$

$= (\mu_0 nI)\pi R^2 = 29{\cdot}6 \times 10^{-6}$ wb

Beginner's Box - 2

1.
(i) Anticlockwise
(ii) Clockwise
(iii) A–Positive charge, B–Negative charge
(iv) Anticlockwise
(v) Anticlockwise
(vi) No induced current
(vii) (a) Anticlockwise
(b) Anti clockwise in bigger loop & clockwise in smaller loop
(c) Anti clockwise in bigger loop & clockwise in smaller loop
(d) Anticlockwise in both loop & through connecting wire zero current
(viii) Anticlockwise

2. (i) (a) Anticlockwise (ACW),
(b) Clockwise(CW)
(ii) N to L
(iii) Plate A – Positive charge,
Plate B – Negative charge

Beginner's Box - 3

1. Induced emf does not depend on nature of coil, it means it doesn't depend on resistance

2. $$e = \frac{d\phi}{dt} = \frac{d\phi}{dx} \times \frac{dx}{dt} = \frac{d\phi}{dx} \times v$$

$e \propto v \quad \left(\text{for } \frac{d\phi}{dx} = \text{constant}\right)$

v_1 = Relative velocity in opposite direction

$= 5+3 = 8$m/s

v_2 = Relative velocity in same direction

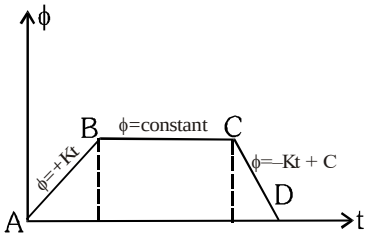
$= 5 - 3 = 2 m/s$

$\frac{e_2}{e_1} = \frac{v_2}{v_1} \Rightarrow e_2 = \frac{2}{8} \times 16 = 4$ mV

3. $e = \frac{-d\phi}{dt} = -\frac{d}{dt}(5t^2 - 4t + 1) = -(10t - 4)$

$= -(10 \times .2 - 4) = -(2 - 4) = 2V$

4.



A toB $\Rightarrow \frac{d\phi}{dt} = +K \Rightarrow e(-K)$

B to C $\Rightarrow \frac{d\phi}{dt} = 0 \Rightarrow e = 0$

C to D $\Rightarrow \frac{d\phi}{dt} = -K \Rightarrow e = +K$

5. $e = -\frac{\Delta\phi}{\Delta t} = \frac{-\Delta(NBA)}{\Delta t} = -NA\left(\frac{\Delta B}{\Delta t}\right)$

$= -N(\pi r^2)\frac{\Delta B}{\Delta t}$

$= -50 \times 3.14 \times (2 \times 10^{-2})^2 \left(\frac{4-2}{3.14}\right)$

$= -50 \times 3.14 \times 4 \times 10^{-4} \frac{2}{3.14} = -0.04$ V

6. $i = \frac{e}{R} = \frac{1}{R}\frac{d\phi}{dt} = \frac{1}{R}A\left(\frac{dB}{dt}\right)\left(\because \phi = BA\right)$

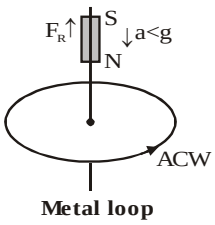
$= \frac{1}{2 \times 10^{-3}} \times 16 \times 10^{-4} \times 0.4 = 0.32A$

7. second equation of motions

$S = ut + \frac{1}{2}gt^2$

$= 0 + \frac{1}{2} \times 9.8 \times 1$

$= 4.9$ m



But acceleration due to induced current a < g

So that distance < 4.9 m

8. induced charge $q = \frac{\Delta\phi}{R} = \frac{10-2}{2} = \frac{8}{2}$

$= 4.0$ C

9. $E(2\pi R) = \pi R^2 \frac{dB}{dt}$

$E = \frac{R}{2}\frac{dB}{dt} = 1 \times 10^{-2} \times 2 = 0.02$ V/m

Beginner's Box - 4

1. $e = -L\left(\frac{\Delta I}{\Delta t}\right) = -5\left(\frac{2-1}{5}\right) = -1.0V$

2. $(I_2 = -2A,\ I_1 = 2A,\ \Delta I = I_2 - I_1)$

$e = -L\frac{\Delta I}{\Delta t} = -2\left(\frac{-2-2}{1}\right) = 8V$

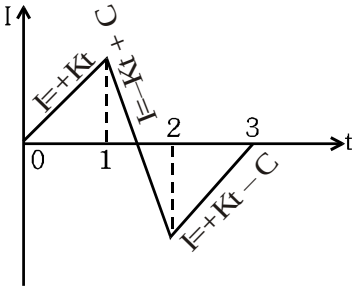
3. $e = -L\frac{dI}{dt}$, if e = 0 then = 0

$2te^{-t} + t^2(-e^{-t}) = 0$

$te^{-t}(2 - t) = 0$

$t = 2$sec.

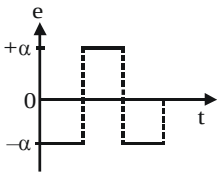
4.



0 to 1, $\frac{dI}{dt} = +K \Rightarrow e = -LK = -\alpha$

1 to 2, $\frac{dI}{dt} = -K \Rightarrow e = +LK = +\alpha$

2 to 3 $\frac{dI}{dt} = +K \Rightarrow e = -LK = -\alpha$



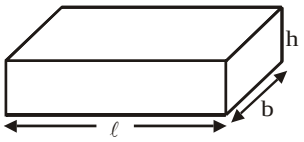
5. $L = \mu_0 n^2 A\ell$

$L \propto \ell$ $(\because$ n & A Constant)

$\frac{L_2}{L_1} = \frac{\ell_2}{\ell_1} \Rightarrow L_2 = \left(\frac{2\ell}{\ell}\right)2 = 4H$

6. $L = \mu_0 n^2 A\ell = \mu_0 n^2 V \Rightarrow L \propto V$

When all linear dimensions are increased by a factor 3, they will become four times



$\frac{L'}{L}=\frac{V'}{V}=\frac{(4\ell)\,(4b)\,(4h)}{\ell\ b\ h}=64\Rightarrow L'=64L$

Increased by factor 63

7. $\phi = LI$

$= 2 \times 5.8 = 11.6$ Wb

Beginner's Box - 5

1. RC and $\frac{L}{R}$, time constant

$\frac{1}{\sqrt{LC}}$ is angular frequency at resonance in a.c.

2. $I = I_0\,(1 - e^{-t/\lambda})$ $\quad\left(\lambda=\frac{L}{R}=\frac{10}{5}=2\text{ sec}\right)$

$=\frac{E}{R}\,(1-e^{-t/\lambda})=1(1-e^{-2/2})=(1-e^{-1})$

3. $\lambda=\frac{L}{R}=\frac{8\times10^{-3}}{4}=2\text{ms}$

5. $\frac{dI}{dt}=\frac{E}{L}(e^{-t/\lambda})$, at t=0 $\frac{dI}{dt}=\frac{E}{L}=\frac{5}{20}=0.25$ A/sec.

6. $\left(\because\lambda=\frac{L}{R}=\frac{5\times10^{-3}}{18}=\frac{.001}{3.6}\right)$

$\frac{dI}{dt}=\frac{E}{L}(e^{-t/\lambda})=\frac{5}{5\times10^{-3}}e^{-\left(\frac{0.001\times18}{5\times10^{-3}}\right)}$

$= 10^3\,(e^{-3.6}) = 1000 \times .0273 = 27.3$ A/sec.

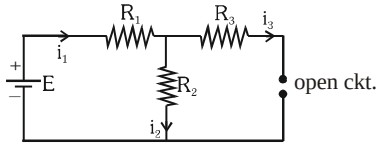
7. $I = I_0/2$ so $t = T = 0.693\ \frac{L}{R}$

$=\frac{0.7\times8.4\times10^{-3}}{6}\simeq1\text{m sec}$

8. $\frac{L}{CVR}=\frac{1}{CV}\times\left(\frac{L}{R}\right)=\frac{\lambda}{Q}\left[\because\lambda=\frac{L}{R},Q=CV\right]$

$\Rightarrow\frac{\text{time}}{\text{charge}}=(\text{current})^{-1}$

9.



At switch on t = 0, L open circuit

$\Rightarrow i_3 = 0$

current $i_1 = i_2 = \frac{E}{R_1+R_2}=\frac{10}{1+2}=\frac{10}{3}=3.3$

$i_1 = i_2 = 3.3$ A, $i_3 = 0$

Beginner's Box - 6

1. $i_2=\frac{e_2}{R_2}\Rightarrow i_2=\frac{1}{R_2}M\left(\frac{dI_1}{dt}\right)$

$\Rightarrow\frac{dI_1}{dt}=\frac{i_2R_2}{M}=\frac{0.4\times5}{0.5}=4\text{ A/s}$

2. $(\phi_2)_T=MI_1\Rightarrow M=\frac{(\phi_2)_T}{I_1}=\frac{9.0\times10^{-5}}{3.0}=3\times10^{-5}$ H

3. $e_2=-\frac{MdI_1}{dt}=-M\frac{d}{dt}(I_0\sin\omega t)=-MI_0\omega\cos\omega t$

4. Peak value $e_0 = MI_0\omega$ (From above question)

$= MI_0(2\pi f)$

$= 1.5 \times 1 \times 2\pi \times 50 = 150\pi$V

5. $M \propto N_1 N_2$ (In question N_1N_2 = constant)

6. $M=\frac{\mu_0N_1N_2A}{\ell}$

$=\frac{4\pi\times10^{-7}\times2000\times1.2\times10^{-3}\times300}{0.3}$

$= 96\,\pi \times 10^{-5}$

$e_m=-M\frac{\Delta I_1}{\Delta t}$

$=-96\pi\times10^{-5}\times\frac{(-4)}{0.25}\left(\begin{array}{l}\because dI=\ I_2-I_1\\ dI=-2-2\\ =-4\end{array}\right)$

$= 4.8 \times 10^{-2}$ V

7. Use trick : $M\propto\frac{x_{in}^2}{x_{out}}$

8. (i) (a) ACW, (b) Zero, (c) CW

(ii) (a) ACW, (b) Zero, (c) CW (seen from right side)

(iii) (a) L to N, (b) Zero, (c) N to L

Beginner's Box - 7

1. (i) No flux cutting $\Rightarrow$ No EMI

(ii) Magnetic field will be upward perpendicular to the plane paper ($\odot$)

(iii) Magnetic field will be downward perpendicular to the plane paper ($\otimes$)

(iv) Q to P

(v) N will be at LP

(vi) X $\rightarrow$ N poleY $\rightarrow$ S pole

(vii) X $\rightarrow$ S poleY $\rightarrow$ N pole

2. $F_{ext}=\frac{B^2\ell^2v}{R}=\frac{0.15\times0.15\times0.5\times0.5\times2}{3}$

$= 3.75 \times 10^{-3}$ N

3. $F_{ext}=\frac{B^2\ell^2v}{R_{loop}}=\frac{0.25\times9\times10^{-4}\times1\times10^{-2}}{1\times10^{-3}}$

$= 2.25 \times 10^{-3}$ N

4. According to Wheatstone bridge resistance of the circuit is 3Ω and resistance of the loop is 1Ω So R_{net} =1+3=4Ω

induced current $i = \frac{B v \ell}{R_{net}} \Rightarrow v = v$

$= \frac{1\times10^{-3}\times4}{2\times10\times10^{-2}}$

$= 2 \times 10^{-2}$ m/s Clockwise (CW)

5. Distance travelled by the loop in t =2s

= (1cm/sec) × 2 sec = 2cm

So that flux of the loop is increasing

$e = B v \ell = 0.6 \times 1 \times 10^{-2} \times 5 \times 10^{-2}$

$= 3 \times 10^{-4}$ V

At t = 10s loop will be completely inside in the magnetic field B

ϕ = contant $\Rightarrow$ e = 0 (No emi)

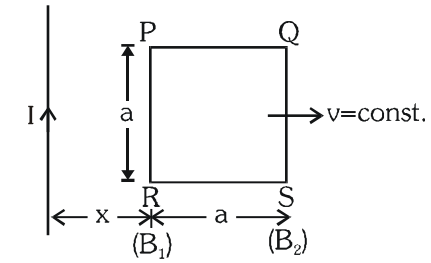
6. $F_{ext} = \frac{B^2\ell^2 v}{R} \Rightarrow F_{ext} \propto$ v straight line

$P_{mech} = \frac{B^2\ell^2 v^2}{R} \Rightarrow P_m \propto v^2$ parabola

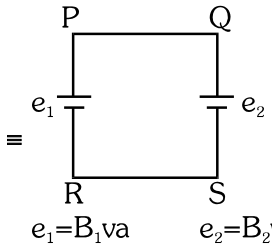
7. Use effective length

$(B \propto \frac{1}{d})$ ⊗ B Non uniform

8.



$(B_1 > B_2)$



$e_1 = B_1 v a$ $e_2 = B_2 v a$

$(e_1 > e_2)$

Instantaneous induced emf in square metal loop is:-

$e_{net} = e_1 - e_2$

$= av(B_1 - B_2)$,

where $B_1 = \frac{\mu_0 I}{2\pi x}$ & $B_2 = \frac{\mu_0 I}{2\pi(x+a)}$

using the values of B_1 and B_2

$e_{net} = \frac{\mu_0 I v a}{2\pi}\left[\frac{1}{x} - \frac{1}{x+a}\right]$

$$\boxed{e_{net} = \frac{\mu_0 I v a^2}{2\pi x(x+a)}}$$ (always non zero)

Beginner's Box - 8

1. e → max ⇒ ϕ → min = 0, When magnetic field is parallel to plane of coil.

2. $e_0 = NBA\,\omega$

$= 60 \times .5 \times 20 \times 10 \times 10^{-4} \times 2 \times \pi \times \frac{1800}{60}$

= 113 V

Beginner's Box - 9

1. To reduce eddy current

2. Efficiency $= \frac{P_{out}}{P_{in}} \times 100 = \frac{11\times90}{220\times5} \times 100 = 90\%$

3. Working of transformer is based on mutual induction

4. Application of eddy current

(i) Induction furnace,

(ii) Electric Brakes

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Magnetic Effect of Current and Magnetism

SOLUTIONS

Beginner's Box - 1

1. According to Biot-Savart's law, the magnetic induction due to a current element is given by:

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{i\vec{d\ell}\times\vec{r}}{r^3}$$

This is perpendicular to both $\vec{d\ell}$ and $\vec{r}$.

2. $B = \frac{\mu_0}{4\pi}\frac{Id\ell\sin\theta}{r^2}$

$$B = \frac{10^{-7}\times10\times1\times10^{-2}}{(0.5)^2}$$

$B = 4 \times 10^{-8}$ T

For direction,

$d\vec{B} = d\vec{\ell}\times\vec{r} = \hat{i}\times\hat{j} = \hat{k}$ i.e +z-direction

3. $B = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2}$

$$B = \frac{\mu_0}{4\pi}\frac{3\times6\pi\sin90^\circ}{3^2}$$

$$B = \frac{\mu_0}{2}\text{Tesla}$$

Beginner's Box - 2

1. $d\vec{B} = \frac{\mu_0}{4\pi}\frac{q(\vec{v}\times\vec{r})}{r^3}$

$d\vec{B}\perp\vec{v}$ & $d\vec{B}\perp\vec{r}$

For direction of magnetic field use right hand thumb rule

Plane of magnetic field and propagation of charge particle are mutually perpendicular.

2. $d\vec{B} = \frac{\mu_0}{4\pi}\frac{q(\vec{v}\times\vec{r})}{r^3}$

$d\vec{B}\perp\vec{v}$ & $d\vec{B}\perp\vec{r}$

For direction of magnetic field use right hand thumb rule

Plane of magnetic field and propagation of charge particle are mutually perpendicular.

3. $d\vec{B}\Rightarrow Id\vec{\ell}\times\vec{r}$

$\Rightarrow \hat{j}\times-\vec{k}$

$\Rightarrow -\hat{i}$ (Toward west)

4. $d\vec{B}\Rightarrow Id\vec{\ell}\times\vec{r}$

$= -\hat{k}\times\hat{i}$

$= -\hat{j}$ (Toward south)

5. Magnetic field at point 'P' due to wire (1) & (2) is zero because point P lies on the axis of these wires.

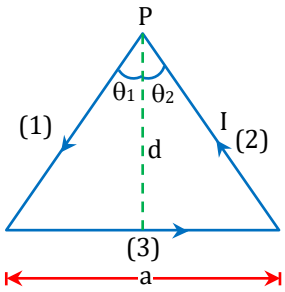
Magnetic field at point P due to wire (3) is given by :-

$B_p = \frac{\mu_0 I}{4\pi d}(\sin\theta_1+\sin\theta_2)$

$d = \frac{\sqrt{3}a}{2}$

and $\theta_1 = \theta_2 = 30^\circ$

$B_P = \frac{\mu_0 I}{4\pi\left(\frac{\sqrt{3}a}{2}\right)}(\sin30^\circ+\sin30^\circ) = \frac{\mu_0 I}{2\sqrt{3}\pi a}\odot$



6. Magnetic field due to wire (1)

$B_1 = \frac{\mu_0(5)}{2\pi(7.5)}\otimes$

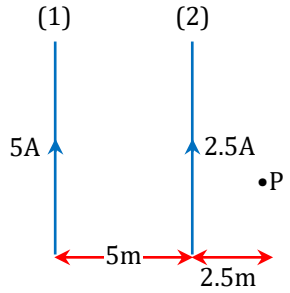
Magnetic field due to wire (2)

$B_2 = \frac{\mu_0(2.5)}{2\pi(2.5)}\otimes$

Net magnetic field at point 'P' due to both wires

$B_p = B_1 + B_2 = \frac{\mu_0(5)}{2\pi(7.5)} + \frac{\mu_0(2.5)}{2\pi(2.5)}$

$= \frac{\mu_0}{2\pi}\left(\frac{2}{3}+1\right) = \frac{5\mu_0}{6\pi}\otimes$



7. Magnetic field due to wire (1) :-

$$B_1 = \frac{\mu_0 I}{4\pi\left(\frac{r}{\sqrt{2}}\right)}(\sin 45° + \sin 90°)$$

$$= \frac{\sqrt{2}\mu_0 I}{4\pi r}\left(\frac{1}{\sqrt{2}} + 1\right)$$

$$= \frac{\sqrt{2}\mu_0 I}{4\pi r}\left(\frac{1+\sqrt{2}}{\sqrt{2}}\right) = \frac{\mu_0 I}{4\pi r}\left(1+\sqrt{2}\right) \otimes$$

Magnetic field due to wire (2) :-

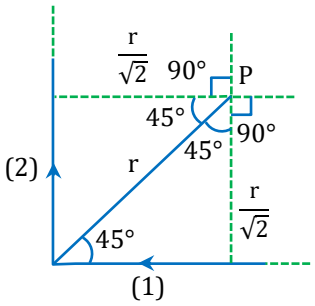
$B_2 = \frac{\mu_0 I}{4\pi r}\left(1+\sqrt{2}\right) \otimes$,

(because point P is symmetrical about both the wires)

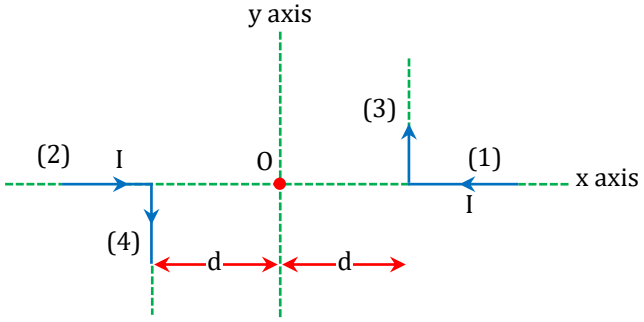
Net magnetic field at point 'P' due to both wires

$$B_p = B_1 + B_2 = 2.\frac{\mu_0 I}{4\pi r}\left(1+\sqrt{2}\right)$$

$$= \frac{\mu_0 I}{2\pi r}\left(1+\sqrt{2}\right) \otimes$$



8. Magnetic field due to wire (1) and (2) is zero because origin lies on the extended axis of these wires



Magnetic field due to wire (3) :-

$$B_3 = \frac{\mu_0 I}{4\pi d}(\sin 0° + \sin 90°) = \frac{\mu_0 I}{4\pi d} \odot$$

Magnetic field due to wire (4)

$$B_4 = \frac{\mu_0 I}{4\pi d}(\sin 0° + \sin 90°) = \frac{\mu_0 I}{4\pi d} \odot$$

Net magnetic field at point 'O' due to wire (3) and (4)

$$B_0 = B_3 + B_4 \quad \frac{\mu_0 I}{4\pi d} + \frac{\mu_0 I}{4\pi d} = \frac{\mu_0 I}{2\pi d} \odot$$

9. Magnetic field due to wire (1) & (2) is zero, because point P lies on the axis of these wires

Magnetic field due to wire (3)

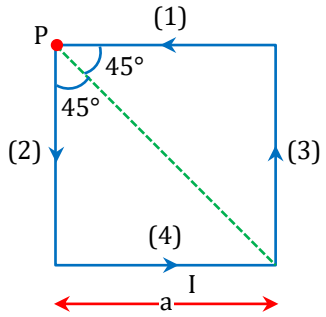
$$B_3 = \frac{\mu_0 I}{4\pi a}(\sin 0° + \sin 45°) = \frac{\mu_0 I}{4\sqrt{2}\pi a} \odot$$

Magnetic field due to wire (4)

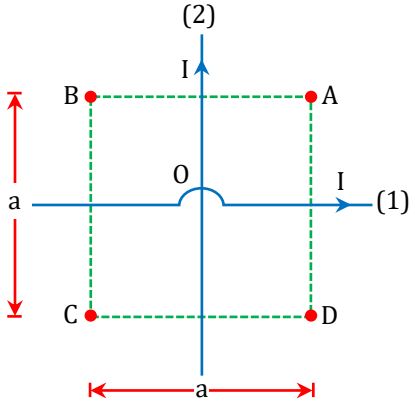
$$B_4 = \frac{\mu_0 I}{4\pi a}(\sin 0° + \sin 45°) = \frac{\mu_0 I}{4\sqrt{2}\pi a} \odot$$

Net magnetic field due to wire (3) and (4) :-

$$B_p = B_3 + B_4 = 2.\frac{\mu_0 I}{4\sqrt{2}\pi a} = \frac{\mu_0 I}{2\sqrt{2}\pi a} \odot$$



10. Direction of magnetic fields at point A and C is opposite but having same magnitude so magnetic field at these points is zero.



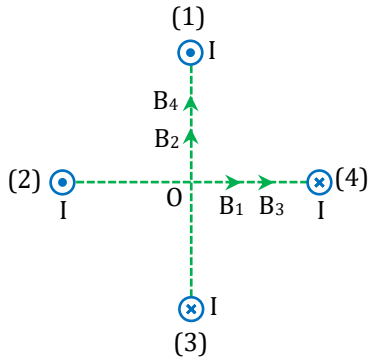
11. $B_1 = B_2 = B_3 = B_4 = B$ (say)

Resultant magnetic field

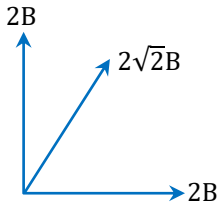
$B_R = \sqrt{(2B)^2 + (2B)^2}$

$= \sqrt{8B^2}$

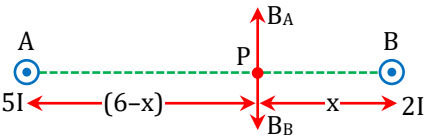
$= 2\sqrt{2}B$



OR



12. (a)



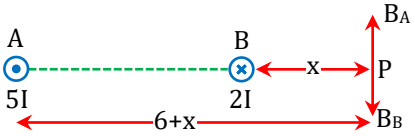
Suppose magnetic field at the distance x from 'B' is zero so at that point magnetic field due to wires A and B will be equal in magnitude.

$B_A = B_B$

$$\frac{\mu_0(5I)}{2\pi(6-x)} = \frac{\mu_0(2I)}{2\pi x}$$

$$\frac{5}{2(6-x)} = \frac{1}{x}\ ; \qquad x = \frac{12}{7}\ \text{cm}$$

(b)



When current is in opposite direction the magnetic field will not be zero between wires A and B. Suppose it is zero at point 'P' then

$B_A = B_B$

$$\frac{\mu_0(5I)}{2\pi(6+x)} = \frac{\mu_0(2I)}{2\pi x}$$

$$\frac{5}{2(6+x)} = \frac{1}{x};\ x = 4\ \text{cm}$$

13. (a) $\tan 60° = \frac{a/2}{d}$

$$d = \frac{a}{2\tan 60°}$$

$$= \frac{a}{2\sqrt{3}}$$

($\because B_1 = B_2 = B_3 = B$ = magnetic field at 'O' due to each side)

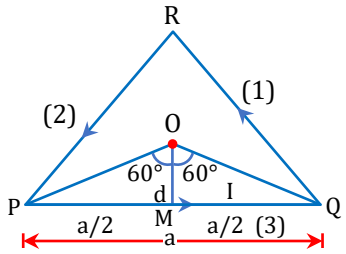
$$B = \frac{\mu_0 I}{4\pi d}(\sin\theta_1 + \sin\theta_2)$$

$$B = \frac{\mu_0 I}{4\pi(a/2\sqrt{3})}(\sin 60° + \sin 60°)$$

$$= \frac{\mu_0 I\sqrt{3}}{2\pi a}\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right) = \frac{3\mu_0 I}{2\pi a}$$

$B_0 = B_1 + B_2 + B_3 = 3B$

$$\therefore\ B_0 = 3B = 3\left(\frac{3\mu_0 I}{2\pi a}\right) = \frac{9\mu_0 I}{2\pi a}\ \odot$$



(b) $\tan 45° = \frac{a/2}{d}$

$$d = \frac{a}{2\tan 45°}$$

$$= \frac{a}{2}$$

($B_1 = B_2 = B_3 = B_4 = B$ = magnetic field at 'O' due to each side)

$$B = \frac{\mu_0 I}{4\pi(a/2)}(\sin 45° + \sin 45°)$$

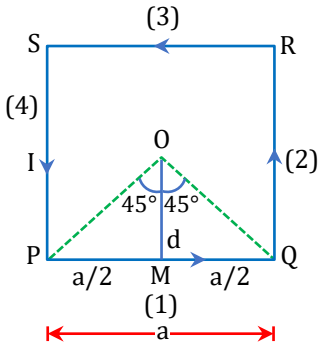
$$B = \frac{\mu_0 I}{4\pi(a/2)}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = \frac{\sqrt{2}\,\mu_0 I}{2\pi a}$$

$B_0 = B_1 + B_2 + B_3 + B_4 = 4B$

$$\therefore B_0 = 4B = 4\left(\frac{\sqrt{2}\,\mu_0 I}{2\pi a}\right) = \frac{2\sqrt{2}\,\mu_0 I}{\pi a}\ \odot$$

OR

Use polygon result, put n = 4



(c) $\tan 30° = \frac{a/2}{d}$

$d = \frac{a}{2\tan 30°} = \frac{\sqrt{3}a}{2}$

($B_1 = B_2 = B_3 = B_4 = B_5 = B_6 = B$ = magnetic field at O due to each side)

$B = \frac{\mu_0 I}{4\pi\left(\frac{\sqrt{3}a}{2}\right)}$ (sin 30° + sin 30°)

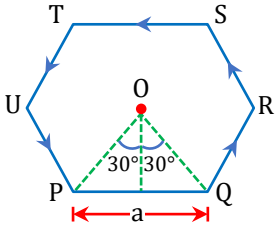
$B = \frac{\mu_0 I}{2\sqrt{3}\pi a}\left(\frac{1}{2}+\frac{1}{2}\right) = \frac{\mu_0 I}{2\sqrt{3}\pi a}$

$B_6 = B_1 + B_2 + B_3 + B_4 + B_5 + B_6 = 6B$

$B_6 = 6\left(\frac{\mu_0 I}{2\sqrt{3}\pi a}\right) = \frac{\sqrt{3}\mu_0 I}{\pi a}$ ⊙

OR

Use polygon result, put n = 6



Beginner's Box - 3

1. Current and number of turns are equal for both coils, so

$B_0 \propto \frac{1}{R} \Rightarrow \frac{B_1}{B_2} = \frac{R_2}{R_1} = \frac{r/2}{r} = \frac{1}{2}$

2. Magnetic field at centre of coil (N=1)

$B_o = \frac{\mu_0 I}{2R}$ ...(1)

From $\ell = 2\pi R$ put $R = \frac{\ell}{2\pi}$ in equation (1)

$B_0 = \frac{\mu_0 I}{2(\ell/2\pi)} = \frac{\mu_0 I\pi}{\ell}$

$B = \frac{4\pi \times 10^{-7} \times 3.14 \times 3.14}{6.28} = 2 \times 10^{-6}$ T

3. (I = constant ℓ = constant)

$B_0 = \frac{\mu_0 NI}{2R}$ $N(2\pi R) = \ell$

$NR = \frac{\ell}{2\pi}$ = constant

$\boxed{B_0 \propto \frac{N}{R}}$ ...(1) $\boxed{N \propto \frac{1}{R}}$ ...(2)

From (1) & (2)

When N given..... $\boxed{B \propto N^2}$

When R given.... $\boxed{B \propto \frac{1}{R^2}}$

According to this question N is given.

$= \frac{B}{B'} = \left(\frac{1}{3}\right)^2$; B' = 9B ; field becomes 9 times

OR

$B_0 = \frac{\mu_0 I}{2R}$ ...(i)

$B'_0 = \frac{\mu_0 NI}{2r}$

$\therefore$ $N \times 2\pi r = 2\pi R$

$r = \frac{R}{N}$

$\therefore$ $B'_0 = \frac{\mu_0 NI.N}{2.R}$

$B'_0 = N^2\left(\frac{\mu_0 I}{2R}\right)$, $\boxed{B'_0 = N^2 B_0}$

here N = 3

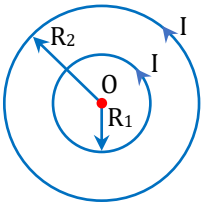
$\therefore$ $B'_0 = 9B_0$

4. Case I :- Current in same direction

$B_i = \frac{\mu_0 NI}{2R_1} + \frac{\mu_0 NI}{2R_2}$

$= \frac{\mu_0 NI}{2}\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$

$= \frac{\mu_0 NI}{2}\left[\frac{R_2+R_1}{R_1 R_2}\right]$ ...(1)



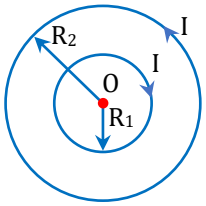
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Case II :- Current in opposite direction

$$B_{ii} = \frac{\mu_0 NI}{2R_1} - \frac{\mu_0 NI}{2R_2}$$

$$= \frac{\mu_0 NI}{2}\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

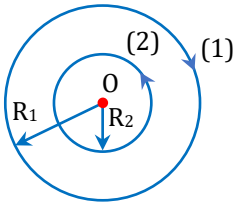
$$= \frac{\mu_0 NI}{2}\left[\frac{R_2 - R_1}{R_1 R_2}\right] \quad ...(2)$$



From (1) & (2)

$$\frac{B_i}{B_{ii}} = \frac{R_2 + R_1}{R_2 - R_1} = \frac{30+10}{30-10} = \frac{2}{1}$$

5. Magnetic field at centre is zero



$B_1 = B_2$

$$\frac{\mu_0 N_1 I_1}{2R_1} = \frac{\mu_0 N_2 I_2}{2R_2}$$

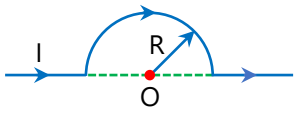
Given that

$(I_1 = I_2, N_1 : N_2 = n_1 : n_2, R_1 : R_2 = 2 : 1)$

$$\frac{N_1}{N_2} = \frac{I_2}{R_2} \times \frac{R_1}{I_1}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{R_1}{R_2} \ (\because I_1 = I_2) \Rightarrow \frac{n_1}{n_2} = \frac{2}{1}$$

6. (a) Magnetic field due to straight parts is zero because point 'O' lies on their extended axis.

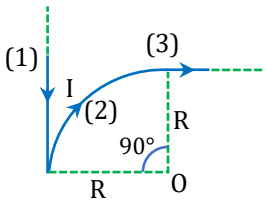


Magnetic field due to circular arc at point 'O' :-

$$B_0 = \frac{\mu_0 I\alpha}{4\pi R} \text{ (Where } \alpha = \pi \text{)};$$

$$B_0 = \frac{\mu_0 I(\pi)}{4\pi R} = \frac{\mu_0 I}{4R} \otimes$$

(b) Magnetic field due to straight wire (1) and (3):-



$$B_1 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°) \odot$$

$$B_3 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°) \otimes$$

Magnetic field due to circular arc (2)

$$B_2 = \frac{\mu_0 I\alpha}{4\pi R},$$

(where $\alpha = \pi/2$)

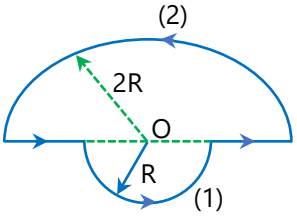
$$B_2 = \frac{\mu_0 I(\pi/2)}{4\pi R} = \frac{\mu_0 I}{8R} = \otimes$$

So net magnetic field at point 'O' :-

$B_0 = B_2 + B_3 - B_1$

$$B_0 = \frac{\mu_0 I}{8R} \otimes$$

7. (a) Point 'O' is located on the extended axis of straight parts so their magnetic field at 'O' will be zero. Magnetic field due to circular arc :-



$$B = \frac{\mu_0 I\alpha}{4\pi R}$$

For arc (1) :-

$$B_1 = \frac{\mu_0 I\pi}{4\pi R} = \frac{\mu_0 I}{4R} \odot \ (\because \alpha = \pi)$$

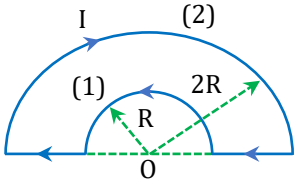
For arc (2) :-

$$B_2 = \frac{\mu_0 I\pi}{4R(2R)} = \frac{\mu_0 I}{8R} \odot \ (\because \alpha = \pi)$$

So net magnetic field at point 'O' :-

$$B_0 = B_1 + B_2 = \frac{\mu_0 I}{4R} + \frac{\mu_0 I}{8R} = \frac{3}{8}\frac{\mu_0 I}{R}$$

(b) Point 'O' is located on the extended axis of straight parts so magnetic field at 'O' will be zero.



For circular arc (1) :-

$B_1 = \frac{\mu_0 I(\pi)}{4\pi R} = \frac{\mu_0 I}{4R}$

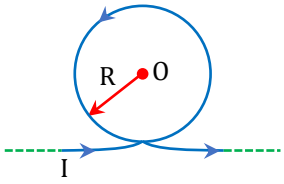
For circular arc (2) :-

$B_2 = \frac{\mu_0 I(\pi)}{4R(2R)} = \frac{\mu_0 I}{8R}$

So net magnetic field at point 'O' :-

$B_0 = B_3 - B_4 = \frac{\mu_0 I}{4R} - \frac{\mu_0 I}{8R} = \frac{\mu_0 I}{8R} \odot$

8. (a) Magnetic field at 'O' due to straight wire :-



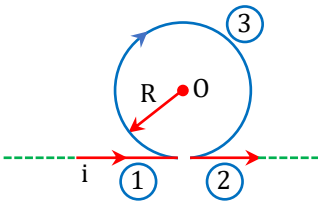
$B_{wire} = \frac{\mu_0 I}{2\pi R} \odot$ and $B_{loop} = \frac{\mu_0 I}{2R} \odot$

So net magnetic field at point 'O' :-

$B_0 = B_{wire} + B_{loop}$

$B_0 = \frac{\mu_0 I}{2\pi R} + \frac{\mu_0 I}{2R} = \frac{\mu_0 I}{2R}\left(\frac{1}{\pi} + 1\right) \odot$

(b) Due to both straight wire B is in same direction



$B_{wire} = B_1 + B_2 \left(B_1 = \frac{\mu_0 I}{4\pi R}, B_2 = \frac{\mu_0 I}{4\pi R}\right)$

$B_{wire} = \frac{\mu_0 I}{2\pi R} \odot$

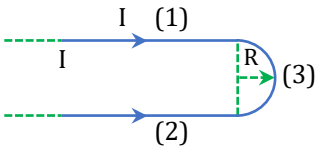
and $B_{loop} = \frac{\mu_0 I}{2R}$

So net magnetic field at point 'O' :-

$B_0 = B_{loop} - B_{wire}$ ($\because$ $B_{loop} > B_{wire}$)

$= \frac{\mu_0 I}{2R} - \frac{\mu_0 I}{2\pi R} = \frac{\mu_0 I}{2R}\left(1 - \frac{1}{\pi}\right) \otimes$

9. (a) Due to wire (1) and (2) magnetic field at 'O' is in same direction so



$B_1 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°) = \frac{\mu_0 I}{4\pi R} = B_2$

Magnetic field due to circular arc :-

$B_3 = \frac{\mu_0 I(\pi)}{4\pi R} = \frac{\mu_0 I}{4R} \otimes$

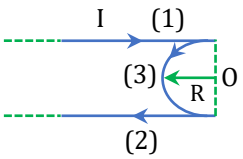
So net magnetic field at point 'O' :-

$B_0 = (B_1 + B_2 + B_3) \otimes$

$B_0 = \frac{\mu_0 I}{4\pi R} + \frac{\mu_0 I}{4\pi R} + \frac{\mu_0 I}{4R} = \frac{\mu_0 I}{2\pi R} + \frac{\mu_0 I}{4R}$

$= \frac{\mu_0 I}{2R}\left(\frac{1}{\pi} + \frac{1}{2}\right) \otimes$

(b) Due to wire (1) and (2) direction and magnitude of field at 'O' is same



So $B_1 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°)$

$= \frac{\mu_0 I}{4\pi R} = B_2 \otimes$

Magnetic field due to circular arc :-

$B_3 = \frac{\mu_0 I(\pi)}{4\pi R} = \frac{\mu_0 I}{4R} \odot$

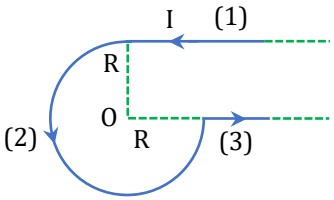
Resultant magnetic field at point 'O' :-

$B_0 = B_3 - (B_1 + B_2)$ [$\because$ $B_3 > (B_1 + B_2)$]

$= \frac{\mu_0 I}{4R} - \frac{\mu_0 I}{2\pi R} = \frac{\mu_0 I}{2R}\left(\frac{1}{2} - \frac{1}{\pi}\right) \odot$

10. (a) Point 'O' is located on the axis of straight part (3) so $B_3 = 0$

Magnetic field due to wire (1)



$$B_1 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°) = \frac{\mu_0 I}{4\pi R}\odot$$

Magnetic field due to circular arc :-

$$B_2 = \frac{\mu_0 I\left(\frac{3}{2}\pi\right)}{4\pi R} = \frac{3\mu_0 I}{8R}\odot$$

So net magnetic field at point 'O' :-

$$B_0 = B_1 + B_2 = \frac{\mu_0 I}{4\pi R} + \frac{3\mu_0 I}{8R}$$

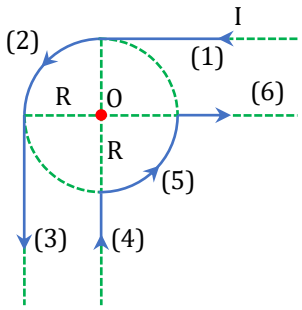
$$= \frac{\mu_0 I}{8R}\left(\frac{2}{\pi} + 3\right)\odot$$

(b) Point 'O' is located on the extended axis of straight parts (4) and (6), so $B_4 = B_6 = 0$

Magnitude and direction of magnetic field at 'O' due to wire (1) and (3) is same so

$$B_1 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°) = \frac{\mu_0 I}{4\pi R} = B_3\odot$$

Due to circular arc (2) and (5) magnetic field at 'O' circular arc (2) and (5) is same so :-



$$B_2 = \frac{\mu_0 I(\pi/2)}{4\pi R} = \frac{\mu_0 I}{8R} = B_5\odot$$

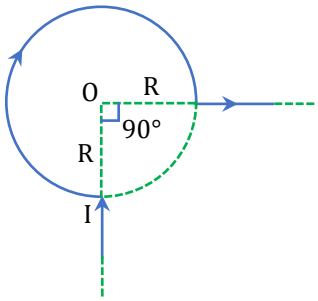
so resultant magnetic field at 'O' :-

$B_0 = B_1 + B_3 + B_2 + B_5$

$$= \frac{\mu_0 I}{4\pi R} + \frac{\mu_0 I}{4\pi R} + \frac{\mu_0 I}{8R} + \frac{\mu_0 I}{8R}$$

$$= \frac{\mu_0 I}{2\pi R} + \frac{\mu_0 I}{4R} = \frac{\mu_0 I}{4R}\left(\frac{2}{\pi} + 1\right)\odot$$

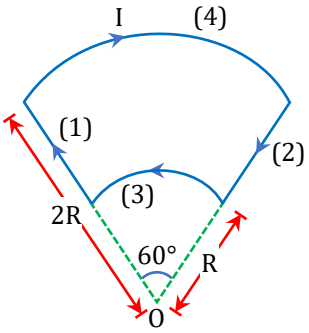
11. (a) Magnetic field due to straight parts is zero because point 'O' lies on their extended axis. Magnetic field due to circular arc



$$B_0 = \frac{\mu_0 I\alpha}{4\pi R}, \text{ (where } \alpha = 3\pi/2)$$

$$B_0 = \frac{\mu_0 I(3\pi/2)}{4\pi R} = \frac{3\mu_0 I}{8R}\otimes$$

(b) Magnetic field due to straight parts (1) and (2) is zero because point 'O' lies on their extended axis.



$B_1 = B_2 = 0$

Magnetic field due to circular arcs (3) and (4)

$$B_3 = \frac{\mu_0 I\alpha}{4\pi R}\odot \text{and}$$

$$B_4 = \frac{\mu_0 I\alpha}{4\pi(2R)}\otimes,$$

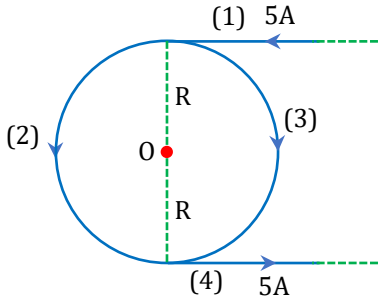
$$(\text{where } \alpha = \frac{\pi}{3})$$

So net magnetic field at point 'O' :-

$$B_0 = (B_3 - B_4) = \frac{\mu_0 I(\pi/3)}{4\pi R}\left[1 - \frac{1}{2}\right] = \frac{\mu_0 I}{24R}\odot$$

Beginner's Box - 4

1. Magnetic field at 'O' due to circular arc (2) and (3) is same in magnitude but direction is opposite. So their net magnetic field is zero.



Due to straight parts (1) and (4) magnitude and direction of magnetic field is same at 'O' so

$B_1 = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°) = B_4 \odot$

net magnetic field at point 'O' :-

$B_0 = B_1 + B_4 = \frac{\mu_0 I}{2\pi R}$

$= \frac{\mu_0 (5)}{2\pi \times 5 \times 10^{-2}} = \frac{50\mu_0}{\pi}$ T$\odot$

$= 2 \times 10^{-5}$ T, $\odot$

2. Magnetic field at the centre due to arc ABC and ADC :-

$B_{ABC} = \frac{\mu_0 I_1}{4r} \otimes$, $B_{ADC} = \frac{\mu_0 I_2}{4r} \odot$

Both arc are connected in parallel with source,

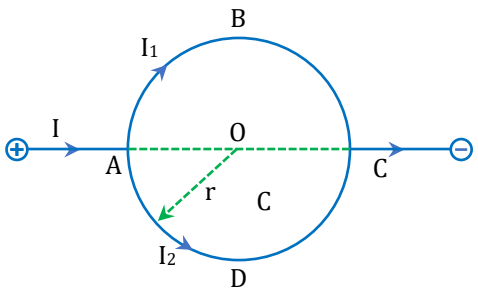
so V= IR = constant

$I_1 = \frac{I(R_2)}{(R_1 + R_2)} = I\left(\frac{10}{15}\right) = \frac{2I}{3}$

$I_2 = \frac{IR_1}{R_1 + R_2} = \frac{I}{3}$

$\therefore$ Net magnetic field at the centre :-

$(B_0)_{net} = \frac{\mu_0}{4r}(I_1 - I_2) = \frac{\mu_0}{4r}\left(\frac{2I}{3} - \frac{I}{3}\right) = \frac{\mu_0 I}{12r} \otimes$



3. $B_{axis} = \frac{\mu_0}{4\pi} \times \frac{2\pi I R^2}{(R^2 + x^2)^{3/2}}$

At centre, $B_{centre} = \frac{\mu_0}{4\pi} \times \frac{2\pi I}{R}$

In the given problem,

$\frac{\mu_0}{4\pi} \times \frac{2\pi I R^2}{(R^2 + x^2)^{3/2}} = \frac{1}{8}\left[\frac{\mu_0}{4\pi} \times \frac{2\pi I}{R}\right]$

or $(R^2 + x^2)^{3/2} = 8R^3$

Solving, we get; $x = \sqrt{3}R$.

4. $B = \frac{B_o}{\left(1 + \frac{x^2}{R^2}\right)^{3/2}}$

$B = \frac{B_0}{\left(1 + \frac{4^2}{3^2}\right)^{3/2}}$

$B = \frac{B_0}{\left(\frac{25}{9}\right)^{3/2}}$

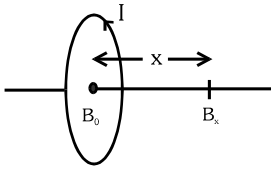
$54 = \frac{27B_0}{125} \Rightarrow B_0 = 250\mu T$

5. $B_x = \frac{\mu_0 N I R^2}{2(r^2 + R^2)^{3/2}}$

If r >> R then $\frac{R^2}{r^2}$ <<< 1 (negligible)

$B_x \simeq \frac{\mu_0 N I R^2}{2(r^2)^{3/2}} \simeq \frac{\mu_0 N I R^2}{2r^3} \Rightarrow B_x \propto \frac{1}{r^3}$

6. Decrease in field value with respect to centre



$\Delta B_x = B_0 - B_x$

$= B_0 - B_0\left(1 - \frac{3}{2}\frac{x^2}{R^2}\right)$

$= B_0\left(\frac{3}{2} \cdot \frac{x^2}{R^2}\right)$

Fractional decrease in field value

(Where x <<< R)

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$$\frac{\Delta B_x}{B_0}=\frac{3}{2}\frac{x^2}{R^2} \quad (\text{limit} \le 5\%)$$

$$\Rightarrow \frac{1}{100}=\frac{3}{2}\frac{x^2}{R^2}$$

$$\Rightarrow x=\pm\frac{R}{\sqrt{150}}$$

Beginner's Box - 5

1. Field inside the conductor

$$B_{in}=\frac{\mu_0 Ir}{2\pi R^2},\ \text{where}\ r=\frac{R}{2}$$

$$=\frac{\mu_0 I(R/2)}{2\pi R^2}=\frac{\mu_0 I}{4\pi R} \quad ...(1)$$

Field outside the conductor :-

$$B_{out}=\frac{\mu_0 I}{2\pi r},$$

$$\text{where } r=2R=\frac{\mu_0 I}{2\pi(2R)}=\frac{\mu_0 I}{4\pi R} \quad ...(2)$$

From (1) & (2)

$$\frac{B_{in}}{B_{out}}=\frac{1}{1}$$

2. Field outside the conductor

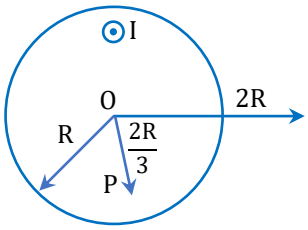
$5\times10^{-6}=\frac{\mu_0 I}{2\pi r}$, where r=2R

$5\times10^{-6}=\frac{\mu_0 I}{2\pi(2R)}$ ------- (1)

Field inside the conductor :-

$$B_p=\frac{\mu_0 Ir}{2\pi R^2},\ \text{where}\ r=\left(R-\frac{R}{3}\right)=\frac{2R}{3}$$

$$=\frac{\mu_0 I(2R/3)}{2\pi R^2}=\frac{1}{3}\left(\frac{\mu_0 I}{\pi R}\right)=\frac{1}{3}(20)\times10^{-6}=\frac{20}{3}\mu T$$



3. Magnetic field at surface of conductor :-

$$B_S=\frac{\mu_0 I}{2\pi R}=B$$

Inside conductor distance r from centre :-

$$B_{in}=\frac{\mu_0 Ir}{2\pi R^2}==\frac{\mu_0 I}{2\pi R}\left(\frac{r}{R}\right)$$

$$\therefore\ B_{in}=B_S\left(\frac{r}{R}\right)$$

$$\frac{B_{in}}{B}=\frac{1}{2},(B_S=B)$$

$$B_{in}=\frac{B}{2}$$

4. (a) Magnetic field inside the conductor :-

$$B_{in}=\frac{\mu_0 Ir}{2\pi R^2}=\frac{\mu_0 r}{2}\left(\frac{I}{\pi R^2}\right)$$

$$=\frac{\mu_0 r}{2}(J),$$

$$(\text{Where } J=\frac{I}{\pi R^2}\ \text{current density})$$

$$=\frac{4\pi\times10^{-7}\times2\times10^{-3}\times100}{2}$$

$$=4\pi\times10^{-8}\ T$$

(b) Magnetic field outside the conductor :-

$$B_{out}=\frac{\mu_0 I}{2\pi r}\ [\because J=\frac{I}{\pi R^2}\Rightarrow I=J(\pi R^2)]$$

$$=\frac{\mu_0 J(\pi R^2)}{2\pi r}=\frac{\mu_0 JR^2}{2r}$$

$$=\frac{4\pi\times10^{-7}\times100\times(3\times10^{-3})^2}{2\times4\times10^{-3}}$$

$$=\frac{9\pi}{2}\times10^{-8}\ T$$

5. By using,

$\oint\vec{B}\cdot\vec{d\ell}=\mu_0\Sigma I$ [Ampere's law]

For loop A : $\oint\vec{B}\cdot\vec{d\ell}=\mu_0(0)=0$

For loop B : $\oint\vec{B}\cdot\vec{d\ell}=\mu_0 I$

For loop C : $\oint\vec{B}\cdot\vec{d\ell}=-\mu_0 I$

For loop D : $\oint\vec{B}\cdot\vec{d\ell}=-\mu_0 I$

6. $\oint\vec{B}.\vec{d\ell}=\mu_0\sum I$

$\sum I=2-1=1A$

$\oint\vec{B}.\vec{d\ell}=\mu_o$

Beginner's Box - 6

2. Magnetic field at axial mid point of solenoid X and Y :–

$B = \mu_0 nI$ where $n = \frac{N}{\ell}$

Magnetic field at axial mid point of new solenoid

$B' = \mu_0 n' I$, (where $n' = \frac{2N}{2\ell} = n$) $= \mu_0 nI = B$

3. As the solenoids are identical, the currents in Q and R will be same and will be half the current in P. The magnetic field within a solenoid is given by $B = \mu_0 nI$. Hence the field in Q will be equal to the field in 'R' and will be half the field in 'P' i.e. will be 1.0T.

4. Magnetic field inside toroid is :-

$$B = \frac{\mu_0 NI}{2\pi R} = 2\left[\frac{\mu_0}{4\pi}\right]\frac{NI}{R}$$

$$B = 2\times10^{-7}\times\frac{240\times2}{20\times10^{-2}} = 48\times10^{-5}$$

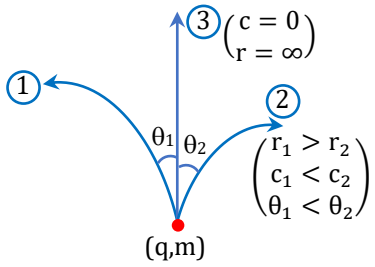
$B = 4.8 \times 10^{-4}$ Tesla

Beginner's Box - 7

1.

	Type	Concept	Ratio
(a)	$v \rightarrow$ ratio	$v \propto \frac{q}{m}$	2 : 1 : 1
(b)	$p \rightarrow$ ratio	$p \propto q$	1 : 1 : 2
(c)	$E_k \rightarrow$ ratio	$E_k \propto \frac{q^2}{m}$	2 : 1 : 2
(d)	$V_{acc.} \rightarrow$ ratio	$V_{acc.} \propto \frac{q}{m}$	2 : 1 : 1

2. **Curvature or sharpness of the curve (C) :–**



$$C \propto \frac{1}{r} \propto \theta$$

Where $r \rightarrow$ radius of curvature or radius
$\theta \rightarrow$ deflection of charge from its initial direction of motion.

	Type q, B → same	Concept $\left(C \propto \frac{1}{r}\right)$	r-ratio ($\theta \propto C$)	Curvature $m_p \simeq 1840\ m_e$	Deflection
(a)	v → same	$r \propto m$ ($r_p > r_e$)	1840 : 1	$C_p < C_e$	$\theta_p < \theta_e$
(b)	p → same	r → same ($r_p = r_e$)	1 : 1	$C_p = C_e$	$\theta_p = \theta_e$
(c)	E_K → same	$r \propto \sqrt{m}$ ($r_p > r_e$)	$\sqrt{1840}$: 1	$C_p < C_e$	$\theta_p < \theta_e$
(d)	$V_{acc.}$ → same	$r \propto \sqrt{m}$ ($r_p > r_e$)	$\sqrt{1840}$: 1	$C_p < C_e$	$\theta_p < \theta_e$

3. (a) $T = \frac{2\pi m}{qB} \Rightarrow T \propto \frac{m}{q}$

$$\Rightarrow T_P : T_d : T_\alpha = \frac{m_p}{e} : \frac{2m_p}{e} : \frac{4m_p}{2e}$$

$$= 1 : 2 : 2$$

(b) $f = \frac{qB}{2\pi m} \Rightarrow f \propto \frac{q}{m}$

$$\Rightarrow f_p : f_d : f_\alpha = \frac{e}{m_p} : \frac{e}{2m_p} : \frac{2e}{4m_p}$$

$$= 1 : \frac{1}{2} : \frac{1}{2}$$

$$= 2 : 1 : 1$$

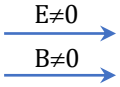
(c) $\omega = \frac{qB}{m} \Rightarrow \omega \propto \frac{q}{m}$

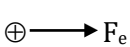
$$\Rightarrow \omega_p : \omega_d : \omega_\alpha = \frac{e}{m_p} : \frac{e}{2m_p} : \frac{2e}{4m_p}$$

$$= 1 : \frac{1}{2} : \frac{1}{2}$$

$$= 2 : 1 : 1$$

4.





v = 0 at t = 0

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Due to electric force charge moves along straight line and parallel to the magnetic field ($\vec{v} || \vec{B}$). So no magnetic force exerts on it during motion.

5. Radius of helical path

$$r = \frac{mv\sin\theta}{qB}$$

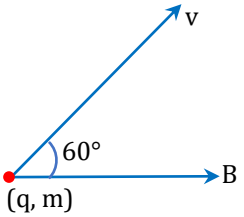
$$= \frac{1.67\times10^{-27}\times4\times10^{5}\times\sqrt{3}/2}{1.6\times10^{-19}\times0.3} = 1.2 \text{ cm}$$

Pitch of helical path :-

$p = (v\cos\theta)T$, [where $T = \frac{2\pi m}{qB}$]

$$= \frac{4\times10^{5}\times\cos60°\times2\times3.14\times1.67\times10^{-27}}{1.6\times10^{-19}\times0.3}$$

≈ 4.35cm



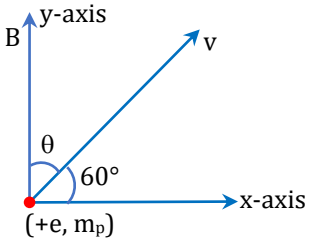
6. Radius of helical path

$r = \frac{mv\sin\theta}{qB}$, where $\theta = (90°-60°) = 30°$

$$= \frac{1.67\times10^{-27}\times2\times10^{6}\times\sin30°}{1.6\times10^{-19}\times0.104} = 0.1\text{m}$$

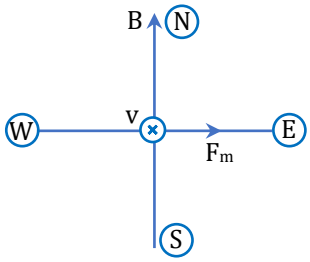
Time period of proton

$$T = \frac{2\pi m}{qB} = \frac{2\times3.14\times1.67\times10^{-27}}{1.6\times10^{-19}\times0.104} = 6.28\times10^{-7}\text{s}$$



7. Time period of circular motion of charge does not depends on angle of projection with the field direction so when angle of projection is 60° then time period remains same.

8. $\vec{v} \perp \vec{B}$, so apply right hand palm rule to find out direction of magnetic force. Hence cosmic particles deflected towards east.



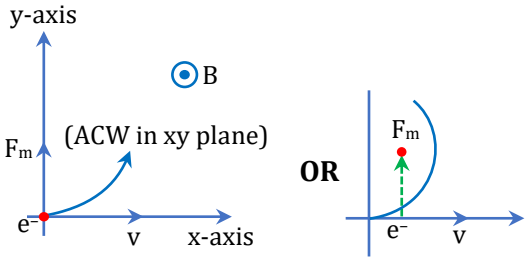
OR

$\vec{F}_m \to (\vec{v}\times\vec{B})$

$\vec{F}_m \to (-\hat{k}\times\hat{j}) = \hat{i}$ east

9. Sense of path gives an idea of direction of magnetic force (i.e. centripetal force), along +y–direction. Now direction of external magnetic field can be find out by right hand palm rule.

Hence magnetic field direction along +z–direction.



Therefore direction of magnetic field is given by right hand palm rule.

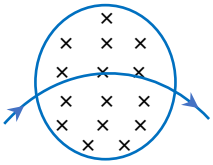
OR

$\vec{F}_m \to (\vec{v}\times\vec{B})$

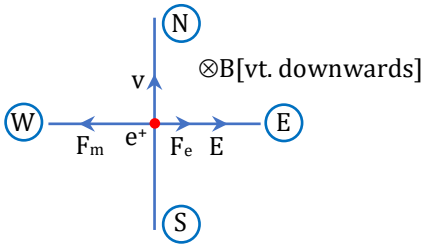
$\hat{j} \to -(\hat{i}\times\hat{k})$

(to obtain the direction of magnetic force along y-axis, magnetic field should be vertically upwards. i.e. along z-axis.

10. According to right hand palm rule positive charges moves on ACW circular path in ⊗ B and negative charges moves on CW circular path in ⊗ B so the path shown in figure is of electron (negative charge)



11. Electric force exerts on proton towards east (in the direction of electric field). For undeflected path magnetic force exerts on proton towards west which is equal to electric force



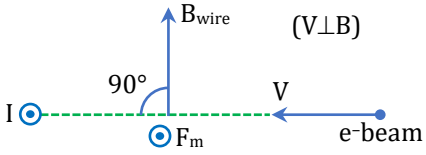
$\vec{F}_e = -\vec{F}_m \Rightarrow |\vec{F}_{net}| = 0, |\vec{a}| = 0 \Rightarrow \vec{v} =$ constant

Now right hand palm rule gives the direction of magnetic field in that region.

12. $\vec{F}_m = \pm q(\vec{v} \times \vec{B})$

$\vec{F}_m \Rightarrow +(\hat{k} + \hat{i}) = +\hat{j}, \Rightarrow$ along + y direction

13. According to right hand palm rule electron beam will be deflected upwards.



OR

$\vec{F}_m \to (\vec{v} \times \vec{B})$

$\vec{F}_m \to -(-\hat{i} \times \hat{j}) = \hat{k}$ (vertically upwards)

14. (a) E = 0, B = 0 $\Rightarrow$ F_{net} = 0 (always)

(b) E = 0, B $\neq$ 0 $\Rightarrow$ F_{net} = 0 (may be when $\vec{v} \,||\, \vec{B}$ or $\vec{v}$ anti $||\, \vec{B}$)

(c) E $\neq$ 0, B $\neq$ 0 $\Rightarrow$ F_{net} = 0 (may be when $\vec{F}_e = -\vec{F}_m$)

(d) E $\neq$ 0, B = 0 $\Rightarrow$ $F_{net} = F_e \neq 0$ (always non zero)

So possible conditions of electric and magnetic field for zero net force on moving charge are (a), (b) and (c)

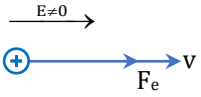
15. $\vec{v}$ = Constant

$|\vec{a}| = 0$

$|\vec{F}_{net}| = 0$ In condition (a), (b), (c) net force on a moving charge may be zero but in (d) $\vec{F}_{net} \neq 0$, $\vec{a} \neq 0 \Rightarrow \vec{v} \neq$ constant (always)

16. In (a),(b),(c) charge may moves along straight line with constant speed (as $\vec{v}$ = may be constant)

In (d) if charge is projected with parallel to electric field ($\vec{v} || \vec{E}$) or anti parallel ($\vec{v}$ anti$||\vec{E}$) then it moves along on straight line with accelerated motion (speed↑)or retarded motion(speed↓)



(θ = 0°) (speed ↑)



(θ = 180°) (speed ↓)

17. (a), (b), (c) $\Rightarrow$ Speed may be constant

(d) $\Rightarrow$ Speed never be constant

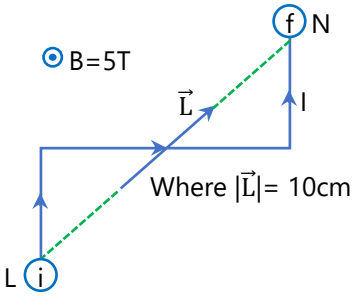
Beginner's Box - 8

1. Magnetic field produced by solenoid is along its axis so magnetic force on conductor given as

$F_m = BIL \sin\theta = (2.5\times10^{-2})\,(5)\,(5\times10^{-2}) \sin 30°$

$= 31.2 \times 10^{-4}$ N

2. $\vec{L}$ and $\vec{B}$ both are perpendicular to each other, so magnetic force on wire 'LN' given



as F_m = BIL $(\vec{L} \perp \vec{B})$

$= 5 \times 10 \times \frac{10}{100} = 5$N

3. Magnetic force on wire ABCDEF in uniform magnetic field given as $\vec{F}_m = I(\vec{L} \times \vec{B})$ Where $\vec{L}$ is the displacement between free ends of the conductor from initial to final point.

$\vec{L} = (\ell)\hat{i}, \vec{B} = (B)\hat{j}$

$\vec{L} \perp \vec{B}$

$\vec{F}_m = BIL(\hat{i} \times \hat{j})$

$= BI\ell\ (\hat{k}) = BI\ell$, along +z direction

4. $F_{PQ} = I\ \ell B\sin\theta = I\ell B\ \sin 90° = I\ell B$ ...(1)

$F_{QR} = I\ell' B\ \sin 30°$

$= I\dfrac{\ell}{\cos 60°} B\sin 30°$

$= I(2\ell)\ B \times \dfrac{1}{2} = I\ell B$ ...(2)

from (1) & (2)

$\dfrac{F_{PQ}}{F_{QR}} = \dfrac{I\ell B}{I\ell B} = 1:1$

OR

In closed loop

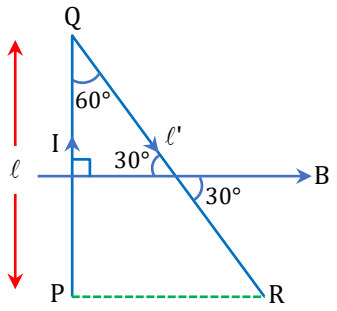
$\vec{F}_{PQ} + \vec{F}_{QR} + \vec{F}_{PR} = 0$

$\vec{F}_{PR} = 0$ (because $\vec{PR} \,||\, \vec{B}$)

$\vec{F}_{PQ} = -\vec{F}_{QR}$

$|\vec{F}_{PQ}| = |\vec{F}_{QR}|$

$\dfrac{F_{PQ}}{F_{QR}} = \dfrac{1}{1}$



5. Magnetic force per unit length

$f_m = \dfrac{\mu_0 I_1 I_2}{2\pi d}$ ...(1)

$f'_m = \dfrac{\mu_0 (2I_1)\ (2I_2)}{2\pi (d/2)} = \dfrac{8\mu_0 I_1 I_2}{2\pi d}$ ...(2)

from (1) & (2)

$f'_m = 8f_m = 8 \times 2 \times 10^{-3} = 16 \times 10^{-3}$ N/m

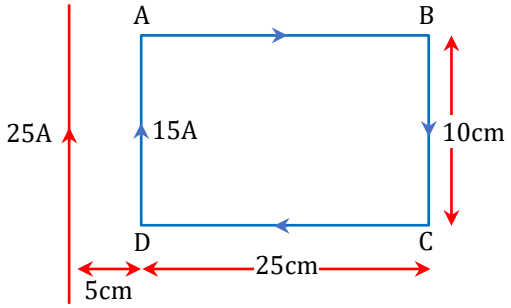
6. Resultant force on 'ℓ' length of wire Q

$F_{net} = (f_R - f_P)\ell$, $(f_R > f_P)$

$= \left[\dfrac{\mu_0 (6)\ (4)}{2\pi (2)} - \dfrac{\mu_0 (2)\ (4)}{2\pi (2)}\right] \times 10 \times 10^{-2}$

$= 16 \times 10^{-8}$ N, towards right

7. Force on 10 cm length of wire AD :-



$F_{AD} = \dfrac{\mu_0 \times 25 \times 15 \times (10 \times 10^{-2})}{2\pi \times 5 \times 10^{-2}}$ ...(1)

$F_{BC} = \dfrac{\mu_0 \times 25 \times 15 \times (10 \times 10^{-2})}{2\pi \times 30 \times 10^{-2}}$ ...(2)

Wire AB and DC located in non uniform magnetic field

So force on elemental length of these wire will be of same magnitude and opposite direction therefore net force on wire AB and DC will be zero $\vec{F}_{AB} + \vec{F}_{CD} = 0$ (always)

$F_{net} = F_{AD} - F_{BC}$ (towards wire) $(F_{AD} > F_{BC})$

$= \dfrac{\mu_0 \times 25 \times 15 \times 10 \times 10^{-2}}{2\pi \times 10^{-2}} \left[\dfrac{1}{5} - \dfrac{1}{30}\right]$

$= 1.25 \times 10^{-4}$ N, attraction

8. Due to unlike currents, magnetic force per unit length of wire 'Q' :-

$f_m = \dfrac{\mu_0 I_1 I_2}{2\pi h}$ N/m ...(1)

Weight per unit length of wire Q :-

$W' = \lambda g$ N/m ...(2)

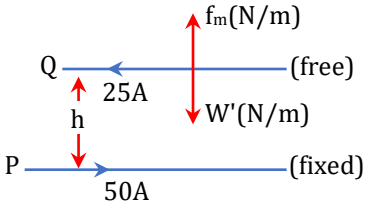
where $\lambda \Rightarrow$ mass per unit length (kg/m)

At balanced condition of wire 'Q' :-

$f_m = W' \Rightarrow \dfrac{\mu_0 I_1 I_2}{2\pi h} = \lambda g$

[from (1) & (2)]

$h = \frac{\mu_0 I_1 I_2}{2\pi\lambda g} = \frac{4\pi\times10^{-7}\times50\times25}{2\pi\times0.075} = 3.33$ mm



9. At balanced condition of free wire :-

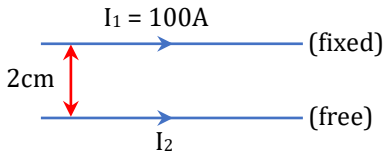
$f_m = \lambda g$

$\frac{\mu_0 I_1 I_2}{2\pi d} = \lambda g$

$I_2 = \frac{2\pi\lambda g d}{\mu_0 I_1}$ ($\lambda = 2\times10^{-3}$ kg/m)

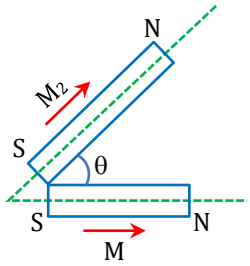
$= \frac{2\times\pi\times2\times10^{-3}\times9.8\times2\times10^{-2}}{4\pi\times10^{-7}\times100}$

= 19.6 A, in free wire



Beginner's Box - 9

1. Magnetic moment of a bar magnet is a vector quantity. So net magnetic moment of following system of two bar magnets of magnetic moments M_1 and M_2 respectively.



$$M_{net} = \sqrt{M_1^2 + M_2^2 + 2M_1M_2\cos\theta}$$

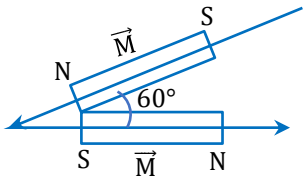
(a) $\theta = 0° \Rightarrow M_{net} = \sqrt{M^2 + M^2 + 2M^2\cos0^\circ}$

= 2M

(b) $\theta = 180° \Rightarrow M_{net} = \sqrt{M^2 + M^2 + 2M^2\cos180^\circ} = 0$

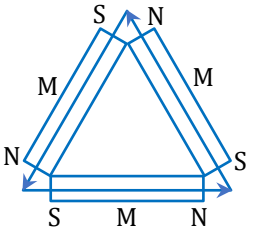
(c) $\theta = 90° \Rightarrow M_{net} = \sqrt{M^2 + M^2 + 2M^2\cos90^\circ} = \sqrt{2}M$

(d)



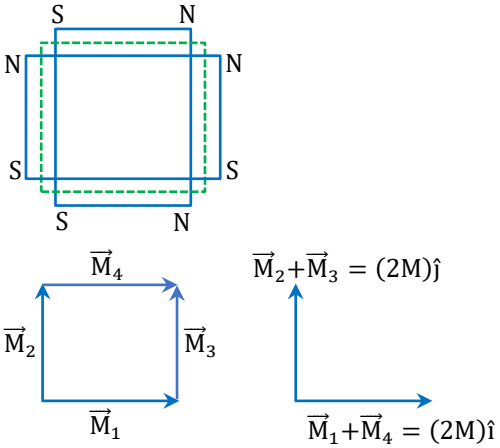
$\theta = 120° \Rightarrow M_{net} = \sqrt{M^2 + M^2 + 2MM\cos120^\circ} = M$

(e)



According to polygon rule of vector addition if all vector can be represented by sides of polygon in sequence the resultant (sum) of all vectors is zero so $M_{net} = 0$

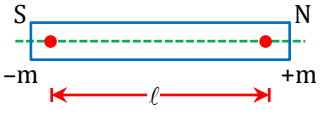
(f)



so total magnetic moment :-

$M_{net} = \sqrt{(2M)^2 + (2M)^2} = 2\sqrt{2}M$

2. Magnetic moment of bar magnet :-



$M = m\ell$

(a) Parallel to its length :-

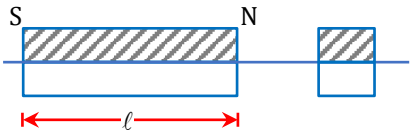
For each part :- $\ell' = \ell$

$m' = \frac{m}{2}$ $\quad (\because A' = \frac{A}{2})$

Magnetic moment of each part :-

$M' = (m/2)\ell$

$$M' = \frac{M}{2}$$



(b) Perpendicular to its length :-

For each part :-

$\ell' = \frac{\ell}{2}$

$m' = m$ $(\because A' = A)$

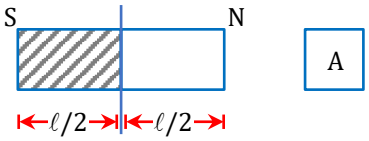
Magnetic moment of each part

$M' = m(\ell/2)$

$$M' = \frac{M}{2}$$

Sp. Note :-

A bar magnet of magnetic moment 'M' cuts into 'n' equal parts then magnetic moment of each part becomes : $M' = \frac{M}{n}$

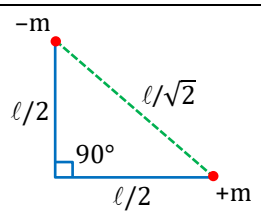
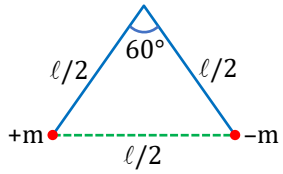
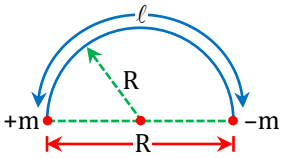
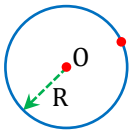


A

3. If a flexible magnet of magnetic moment 'M' is bent into any shape then magnetic moment of new shape always decreases as effective length always decreases (but pole strength remains same)



$\Rightarrow M = m\ell$

a		$M' = m\left(\ell/\sqrt{2}\right)$ $M' = M/\sqrt{2}$
B		$M' = m(\ell/2)$ $M' = \frac{M}{2}$
C		$M' = m(2R)$, where $\ell = \pi R$ $M' = 2m(\ell/\pi)$, $M' = \left(\frac{2M}{\pi}\right)$
d		Effective length of system is zero so magnetic moment is also zero.

4. Magnetic moment of current carrying loop given as $M = IA$

(a) $A_T = \frac{\sqrt{3}a^2}{4} \Rightarrow M_T = I\left(\frac{\sqrt{3}a^2}{4}\right)$

$$\Rightarrow M_T = \frac{\sqrt{3}I\ell^2}{36} \quad \text{(where } \ell = 3a\text{)}$$

(b) $A_S = a^2 \Rightarrow M_S = I(a^2)$

$$\Rightarrow M_S = \frac{I\ell^2}{16} \quad \text{(where } \ell = 4a\text{)}$$

(c) $A_H = 6\left(\frac{\sqrt{3}\,a^2}{4}\right) \Rightarrow M_H = I\,\frac{3\sqrt{3}\,a^2}{2}$

$$\Rightarrow M_C = \frac{\sqrt{3}I\ell^2}{24} \quad \text{(where } \ell = 6a\text{)}$$

(d) $A_C = \pi R^2 \Rightarrow M_C = I(\pi R^2)$

$$\Rightarrow M_C = \frac{I\ell^2}{4\pi} \quad \text{(where } \ell = 2\pi R\text{)}$$

5. I = Constant ℓ = Constant

$N(2\pi R) = \ell$

$R \propto \frac{1}{N}$ ...(1)

Magnetic moment of coil

$M = NI(\pi R^2)$

$M \propto NR^2$ ...(2)

From (1) & (2)

$N \rightarrow$ given $\Rightarrow \boxed{M \propto \frac{1}{N}}$

$R \rightarrow$ given $\Rightarrow \boxed{M \propto R}$

So $\frac{M_1}{M_2} = \frac{N_2}{N_1} = \frac{2}{1}$

6. From Q.No. (6) $\boxed{M \propto \frac{1}{N}} \Rightarrow N_{min} = 1 \Rightarrow M_{max}$

7. Magnetic moment is an axial vector and axis of both the coils are perpendicular to each other. So their magnetic moments also perpendicular to each other.

$M_{net} = \sqrt{(M)^2 + (M)^2} = \sqrt{2}M$

Beginner's Box - 10

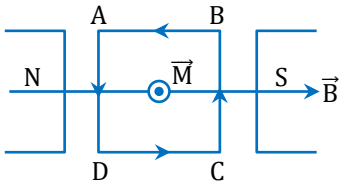
1. In non-uniform magnetic field. $F \neq 0$ (always) and $\tau = 0$ (may) or $\tau \neq 0$ (may) so needle may experiences force and torque.

2. $\vec{\tau} = \vec{M} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 50 & 0 & 0 \\ 0.5 & 3 & 0 \end{vmatrix}$

$= [(0)\hat{i} - (0)\hat{j} + (150)\hat{k}]$ N–m $= 150\ \hat{k}$ N–m

3. $\tau = BINA$

$= 0.2 \times 2 \times 100 \times \frac{8}{100} \times \frac{10}{100} = 0.32$ N–m



$(\vec{M} \perp \vec{B}, \theta = 90°)$

Tendency of torque is try to align $\vec{M}$ in the direction of ext. field $[\vec{\tau} = \vec{M} \times \vec{B}]$. So AD side comes out from plane of paper.

4. For the given perimeter circular shape has maximum geometrical area.

So $A_{max} \Rightarrow M_{max} \Rightarrow \tau_{max} \Rightarrow$ couple max.

5. For constant length wire Torque on coil

$\boxed{M \propto \frac{1}{N}}$ ...(1) $\boxed{\tau \propto M}$ ...(2)

So $\tau_{max} \Rightarrow M_{max} \Rightarrow N_{min} = 1$ (one turn coil)

6. Work done in rotating a dipole from θ_1 to θ_2 from the field direction :-

$W = MB(\cos\theta_1 - \cos\theta_2)$

$= MB(\cos 30° - \cos 45°)$

$= 4 \times 5 \times 10^{-4}\left(\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\right) = 3.2 \times 10^{-4}$ J

7. Work done in rotating a dipole from $\theta_1 = 0°$ to $\theta_2 = 60°$ from the field direction :-

$W = MB(1 - \cos 60°)$

$= \frac{MB}{2}$ ...(1)

The torque needed to maintain the needle in this position

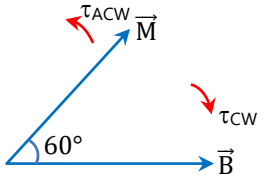
$\tau_{ACW} = \tau_{CW}$

$= MB \sin 60°$

$= \frac{\sqrt{3}MB}{2}$ ...(2)

from (1) & (2)

$\tau = \sqrt{3}\ W$



8. $\tau = MB(\cos\theta_1 - \cos\theta_2)$

$= 2.5 \times 0.2\ (\cos 0° - \cos 180°)$

$= 0.5\ [1 - (-1)] = 1$J

9. $\tau = MB(\cos\theta_1 - \cos\theta_2) = NI(\pi R^2)B(\cos 0° - \cos 90°)$

$= NI\pi R^2 B$

10. $\tau = MB\sin\theta$

$\Rightarrow MB = \frac{\tau}{\sin\theta} = \frac{0.032}{\sin 30°} = 0.064$ N-m or J

(a) $U = -MB\cos\theta$ [In stable equilibrium $\vec{M} \parallel \vec{B}$ $(\theta = 0°)$] $= -0.064$J

(b) $U = -MB\cos\theta$ [In unstable equilibrium $\vec{M}$ antiparallel to $\vec{B}$ $(\theta = 180°)$] $= +0.064$J

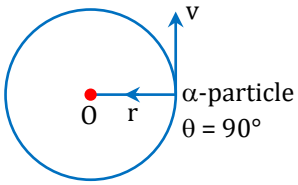
Beginner's Box - 11

1. $B_0 = \frac{\mu_0}{4\pi}\frac{qv\sin\theta}{r^2}$

Charge on α-particle = 2e

$\therefore\ B_0 = \frac{\mu_0}{4\pi}\frac{(2e)v\sin 90^\circ}{r^2}$

$= \frac{\mu_0}{2\pi}\frac{ev}{r^2}$



2. Magnetic field $\vec{B}$ due to moving charge is given as

$\vec{B} = \frac{\mu_0 q(\vec{v}\times\vec{r})}{4\pi r^3}$

$B = \frac{\mu_0 qvr}{4\pi r^3}$

$v = \frac{\text{distance}}{\text{Time}} = 2\pi rn\ \text{m/s}$

$B = \frac{\mu_0 q}{4\pi r^2}\times 2\pi rn$

$B = \frac{\mu_0 qn}{2r}$

Beginner's Box - 12

1. $\frac{B_{axis}}{B_{eq.}} \simeq \frac{2KM/d^3}{KM/d^3} \simeq \frac{2}{1}$

2. $B_{axis} = B_{eq.} \Rightarrow \frac{\mu_0}{4\pi}\frac{2M}{x^3} = \frac{\mu_0}{4\pi}\frac{M}{y^3}$

$\frac{2}{x^3} = \frac{1}{y^3} \Rightarrow \frac{x^3}{y^3} = \frac{2}{1} \Rightarrow \frac{x}{y} = 2^{1/3}$

3. Magnetic lines inside bar magnet is from S→N and outside N→S.

4. $B_P = \frac{\mu_0}{4\pi}\frac{2M}{r^3}, B_Q = \frac{\mu_0}{4\pi}\frac{M}{(2r)^3}, B_R = \frac{\mu_0}{4\pi}\frac{2M}{(4r)^3}$

$B_P : B_Q : B_R = 2 : \frac{1}{8} : \frac{2}{64}$

or 64 : 4 : 1

Beginner's Box - 13

4. For solenoid B = μni

$4\pi = \mu \times 10^3 \times 2$

$2\pi \times 10^{-3} = \mu$

Now $\mu_r = \frac{\mu}{\mu_0} = \frac{2\pi\times10^{-3}}{4\pi\times10^{-7}}$

$\mu_r = 5 \times 10^3$

$\chi = \mu_r - 1;\ \chi = 4999$

5. Flux $\phi = BA$

$\phi = (\mu HA)$

$3.14 \times 10^{-5} = \mu\ 1600 \times 0.2 \times 10^{-4}$

$\mu = 9.8 \times 10^{-4}$

$\therefore\ \mu_r = \frac{\mu}{\mu_0} = \frac{9.8\times10^{-4}}{4\pi\times10^{-7}}$

$\mu_r = 780.25$

$\therefore\ \chi = \mu_r - 1 = 779.25$

8. $\phi = BA$

$4 \times 10^{-4} = \mu \times 1600 \times 4 \times 10^{-4}$

$\mu = \frac{1}{1600}$; $\mu = 0.625 \times 10^{-3}$ Wb/A-m

Capacitor

SOLUTIONS

BEGINNER'S BOX-1

1. $\phi = \frac{q_{enc}}{\varepsilon_0} \because q_{enc} = 0 \therefore \phi = 0$

2. Zero, $\frac{1}{2}\frac{Q^2}{C}$

3. $U \propto Q^2$

$\frac{U_1}{U_2} = \left(\frac{Q_1}{Q_2}\right)^2 \Rightarrow \frac{U_1}{U_2} = \left(\frac{1}{2}\right)^2$

$\Rightarrow U_2 = 4W$

4. $R = n^{1/3}r$

$R = 8^{1/3} r$

$C' = 4\pi\in_0 R = 2(4\pi\in_0 r) = 2C$

5. $C = \frac{4\pi\in_0 b^2}{b-a}$

Here $C = 1\mu F$, $b - a = 1$ mm

$\Rightarrow 1\times10^{-6} = \frac{1}{9\times10^9}\frac{b^2}{1\times10^{-3}}$

On solving we get b = 3 m

6. $C = \frac{Q}{V} = \frac{2\times10^{16}\times1.6\times10^{-19}}{10}$

$= 3.2 \times 10^{-4}$ F

7. A Because $C = \frac{Q}{V} \propto \frac{1}{\text{slope}}$

$\Rightarrow C_A > C_B$

8. $Q = CV = 0.09$ C

$\Rightarrow U = \frac{Q^2}{2C} = 135$ J

9. According to question

$\frac{1}{2}C_1V_1^2 = \frac{1}{2}C_2V_2^2$

$\frac{V_1}{V_2} = \sqrt{\frac{C_2}{C_1}}$

BEGINNER'S BOX-2

1. Work = $U_F - U_i$

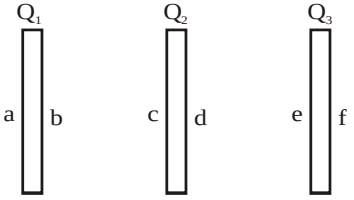
and charge remains constant

$\because C = \frac{\varepsilon_0 A}{d}$ $C' = \frac{\varepsilon_0 A}{2d}$

$V = V$ $V' = 2V$

$\therefore U_i = \frac{\varepsilon_0 AV^2}{2d}$ $U_F = \frac{\varepsilon_0 A(2V)^2}{2(2d)}$

2.



Changes on outer faces = $\frac{Q_1+Q_2+Q_3}{2}$

3. $C_{medium} = KC$

$\Rightarrow K = \frac{175pF}{50pF} = 3.5$

4. $C = \frac{\in_0\in_r A}{d}$; $U = \frac{1}{2}CV^2$

$= \frac{1}{2} \times (5.94 \times 10^{-11}) \times (12)^2 = 4.3 \times 10^{-9}$ J

5. $C = \frac{\in_0 A}{d - t + \frac{t}{K}}$ $\left(t = \frac{d}{2}, K = \infty\right)$

$= \frac{\in_0 A}{\left(d - \frac{d}{2}\right) + \frac{d}{2K}} = \frac{2\in_0 A}{d} = 2C_0$

6. $C' = KC_0 = 5C_0$

$U' = \frac{Q^2}{2C'} = \frac{W_0}{5}$

7. $\because$ Electric field $E = \frac{V}{d}$

$\therefore d = \frac{V}{E} = \frac{10^3}{10^6} = 10^{-3}$ m

Capacitance $C = \frac{\in_0\in_r A}{d}$

$\Rightarrow A = \frac{Cd}{\in_0\in_r} = \frac{88.5\times10^{-12}\times10^{-3}}{8.85\times10^{-12}\times10} = 10^{-3}$ m^2

8. (a) Charge on C $Q = CV = 10\times 20\ \mu C$

$= 200\ \mu C$

(b) Charge on C' $Q' = C'V = 10\times 4\times 20$

$= 800\ \mu C$

9. Since battery is connected so voltage across capacitor remains constant

Initial charge on capacitor

$Q = CV = 10 \times 10^{-6} \times 10 = 100\mu C$

When liquid of dielectric constant 5 is filled between plates then capacity becomes

$C' = 5C = 50\mu F$

So final charge after placing liquid

$Q' = C'V = 50 \times 10^{-6} \times 10 = 500\mu C$

So additional charge flowing from battery to capacitor $\Delta Q = Q' - Q = 500 - 100 = 400\mu C$

10. Energy density $= \frac{1}{2}\in_0 E^2$

$\because \quad E = \frac{V}{d}$

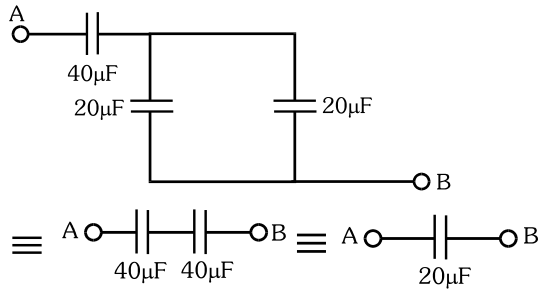
So energy density $= \frac{1}{2}\in_0 \frac{V^2}{d^2}$

11. $\sigma_{big} = (n)^{\frac{1}{3}} \times \sigma_{small} = (64)^{\frac{1}{3}} \times \sigma_{small}$

$\sigma_{big} = 4 \times \sigma_{small}$

BEGINNER'S BOX-3

1. Given circuit can be reduced to



Therefore $C_{AB} = 20\ \mu F$

If $V_{AB} = 100$ V then charge flow through circuit

$= C_{AB}V_{AB} = (100)(20) = 2 \times 10^{-3}$ C

2. (a) Net capacitance $C_{net} = \frac{C}{3} = \frac{9}{3} = 3pF$

(b) in series $V = V_1 + V_2 + V_3 = 3V_1$

($\because V_1 = V_2 = V_3$ identical capacitor)

Potential difference across each

$V_1 = \frac{V}{3} = \frac{120}{3} = 40V$

3. Common potential $V_{cm} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

$= \frac{2\times200 + 3\times400}{2+3} = 320$ V

Charge on $C_1 = Q_1 = C_1V_{cm} = 2 \times 320\ \mu C$

$= 640\ \mu C$

Charge on $C_2\ Q_2 = C_2V_{cm} = 3 \times 320\ \mu C$

$= 960\ \mu C$

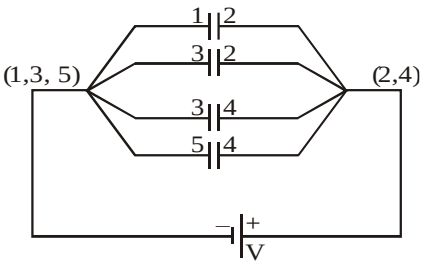
4. $C_{eff} = \frac{C_1C_2}{C_1 + C_2}$

$C_1 = \frac{K_1 \in_0 A}{d/2} = \frac{2K_1 \in_0 A}{d}$

$C_2 = 2K_2\frac{\in_0 A}{d}$

$C_{eff} = \frac{2K_1K_2}{(K_1 + K_2)}\frac{\in_0 A}{d}$

5.

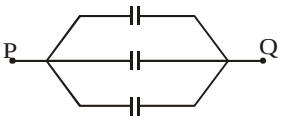


$Q = CV = \frac{\in_0 A}{d}V$

Charge on plate $1 = -\frac{\in_0 A}{d}V$

Charge on plate $4 = \frac{\in_0 AV}{d} + \frac{\in_0 A}{d}V = 2\frac{\in_0 A}{d}V$

6.



$C_{PQ} = \frac{3\in_0 A}{d}$

7. (a) Potential energy $U_{each} = \frac{1}{2}CV^2$

(b) Common potential $V_{cm} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

$= \frac{CV - CV}{C + C} = 0$

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(c) Charge $Q'_1 = C_1V_{cm} = C(0) = 0$ &
$Q'_2 = C_2V_{cm} = C(0) = 0$

(d) Potential energy
$U'_1 = \frac{1}{2}C_1V^2_{cm} = 0$ & $U'_2 = \frac{1}{2}C_1V^2_{cm} = 0$

(e) Potential energy loss = $U_i - U_f$
$= 2 \times \frac{1}{2}CV^2 - 0 = CV^2$

BEGINNER'S BOX-4

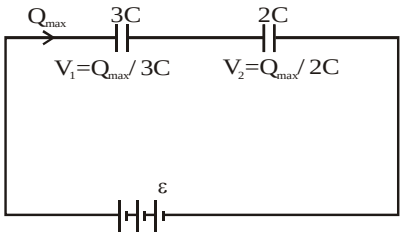
1. (i) For capacitor
$\frac{V_0}{R} = 10$
$\Rightarrow V_0 = 10R = 100$ volt

(ii) $i = 10e^{-t/RC}$
$\Rightarrow 2.5 = 10e^{-2/RC}$
$\Rightarrow RC = \tau = \frac{1}{\ell n2}$ & $C = \frac{1}{10\ell n2}$

(iii) Total heat developed
= Total initial energy stored in capacitor.
$= \frac{1}{2}CV^2 = \frac{500}{\ell n2}$

(iv) Time - constant = $RC = \frac{1}{\ell n2}$

2.

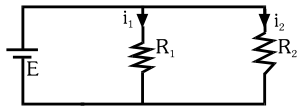


$q_{1max} = 3C(1\text{ kV})$; $q_{2max} = 2C(2\text{ kV})$
$\Rightarrow Q_{max} = q_1$
So, $V_1 = 1$ kV ; $V_2 = 1.5$ kV
Safe voltages in each arm are mentioned.
$E_{safe} = 1+1.5 = 2.5$ kV

3. V decreases continuously from left to right except in conductor where it is constant.

4. Total charge $Q = CV = 400 \times 100$
$= 40000\ \mu C = .04$ C
time taken = $\frac{\text{Total charge}}{\text{Charge rate}}$
$= \frac{0.04}{100\times10^{-6}} = 400$ s

5. (a)



At t = 0, capacitor has zero resistance, i.e., R_1
and R_2 are in parallel. The simple circuit is shown in fig.
$i_1 = \frac{E}{R_1}$ and $i_2 = \frac{E}{R_2}$

(b) At steady state ($t=\infty$), capacitor has infinite resistance. Hence, $i_1 = \frac{E}{R_1}$, $i_2 = 0$

(c) Final potential difference across capacitor is E. $\therefore$ Final energy stored
$U = \frac{1}{2}CE^2$

(d) When switch is opened, capacitor will discharge through two resistance as R_1 and R_2 (both in series).
Hence, time constant $\tau_c = C(R_1 + R_2)$

(e) When switch is closed, capacitor will charged through resistance R_2.
So $\tau = R_2C$

6. $H = \frac{1}{2}CV^2 = \frac{1}{2}\times2\times10^{-6}\times200\times200 = 0.04$ J

Current Electricity

SOLUTIONS

BEGINNER'S BOX-1

1. $I = \frac{Q}{t} = \frac{10000\times2\times1.6\times10^{-19}}{60}$

$= 0.5 \times 10^{-16}$ A

2. $I = \frac{Ne}{t} = ne \Rightarrow n = \frac{I}{e} = \frac{3.2\times10^{-3}}{1.6\times10^{-19}} = 2 \times 10^{16}$

3. $v_d = \frac{I}{neA}$

$= \frac{6.25}{8.5\times10^{28}\times1.6\times10^{-19}\times3.14\times(1\times10^{-3})^2}$

$= 0.15$ mm/s

4. $I = \frac{Q}{t} = \frac{90}{4500} = 0.02$ A

$v_d = \frac{I}{nAe}$

$= \frac{0.02}{(5.8\times10^{28}\times10^{+6})(1.6\times10^{-19})\left[\frac{\pi}{4}\times(10^{-3})^2\right]}$

$= 2.74 \times 10^{-12}$ m/s

BEGINNER'S BOX-2

1. Resistance $R = \frac{\rho\ell}{A}$ where $\ell = 100$ m,

$A = 1 \times 1 = 1m^2$

$\therefore R = \frac{4\times10^{-6}\times100}{1} = 4 \times 10^{-4}\ \Omega$

2. $R = \frac{\rho\ell}{A} = \frac{\rho(A\ell)}{A^2} \Rightarrow R \propto \frac{1}{A^2} \propto \frac{1}{r^4}$

$\frac{\Delta R}{R}\times100 = 4 \times \frac{\Delta r}{r}\times100 = 4 \times 0.1 = 0.4\%$

3. $\because$ Resistance $R = \rho\frac{\ell}{A} = \rho\frac{\ell.\ell}{A.\ell} = \rho\frac{\ell^2}{V}$

($\because$ ρ & V are constant)

$R_2 = R_1\left(\frac{\ell_2}{\ell_1}\right)^2 = R_1\left(\frac{2\ell_1}{\ell_1}\right)^2 = 4R_1 = 4 \times 5 = 20\ \Omega$

Resistivity of the wire remains unchanged as it does not change with change in dimensions of a material without change in its temperature.

4. $\because$ $R_\theta = R_0(1 + \alpha\theta)$ where $\alpha = \frac{1}{273}\ ^\circ C^{-1}$

& $R_\theta = 2R_0$

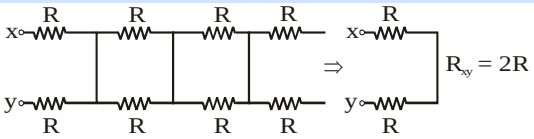
$\Rightarrow 2R_0 = R_0(1 + \alpha\theta) \Rightarrow \theta = \frac{1}{\alpha} = 273^\circ C$

5. (A) – q, (B) – p, (C) – r

6. (A) – s, (B) – p, (C) – q,t, (D) – r

BEGINNER'S BOX-3

1.



2.



$R_{AB} = nr + r/n$

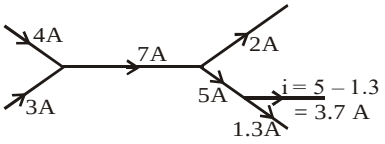
$= \frac{r(n^2+1)}{n}$

3. Connect them in parallel combination

4. $\frac{R_1}{R_2} = \frac{A_2}{A_1} \Rightarrow \frac{R_1}{10} = \frac{3}{1} \Rightarrow R_1 = 30\ \Omega$

$R_{eq} = R_1 + R_2 = 40\ \Omega$

5.



6. $i = \frac{12-8}{12} = \frac{1}{3}$ A

$V_{PQ} = 9 \times \frac{1}{3} = 3$ Volt

BEGINNER'S BOX-4

1. $i = \frac{220-(1.8\times20)}{R+(0.1\times20)} = 15 \Rightarrow R = 10.27\ \Omega$

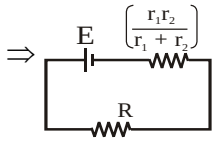
2. When i = 0 V = E = 2 Volts

$r = \frac{E-V}{i} = \frac{2}{5} = 0.4\ \Omega$

3. $i = \frac{8-4}{4.5+0.5+1+2} = 0.5$ A

$E_1 = 4 + (0.5 \times 0.5) = 4.25$ Volts

$E_2 = 8 - (0.5 \times 1) = 7.5$ Volts

4. $\Rightarrow$ 

For max power in R $\Rightarrow$ R = $\frac{r_1r_2}{r_1+r_2}$

5. $\frac{V^2}{R} = P \Rightarrow V = \sqrt{R \times P} = \sqrt{196 \times 1} = 14 \text{ volt}$

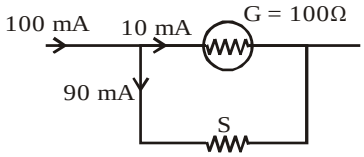
6. Let resistances of bulbs are R and 2R

Power dissipated $P = \frac{V^2}{R}$

$\therefore P_1 = \frac{V^2}{R}$

and $P_2 = \frac{V^2}{2R}$ $\quad \therefore \frac{P_1}{P_2} = \frac{2}{1}$

BEGINNER'S BOX-5

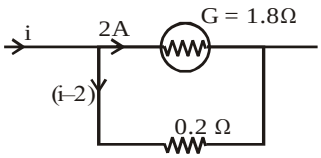
1. 

10 mA × 100Ω = S × 90 mA

S = 11.11 Ω

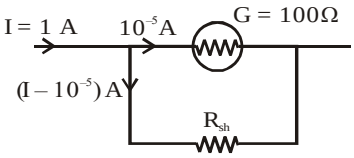
2. S = (n – 1) Ghere n = $\frac{20}{20 \times 0.01} = 100$

S = 99 × 20 = 1980Ω = 1980 Ω

3. 

(i – 2) × 0.2 = 2 × 1.8 $\Rightarrow$ i = 20 A

4. G = 100 Ω, $I_g = 10^{-5}$ A



$10^{-5} \times 100 = (I - 10^{-5})R_{sh}$

$10^{-3} = (1 - 10^{-5})R_{sh} \Rightarrow R_{sh} \approx 10^{-3}\ \Omega$

5. $R_s = \frac{R_g \times i_g}{i - i_g}$

here $i_g = \frac{10}{100} \times i = \frac{i}{10} \Rightarrow R_s = \frac{99 \times \frac{i}{10}}{i - \frac{i}{10}} = 11\ \Omega$

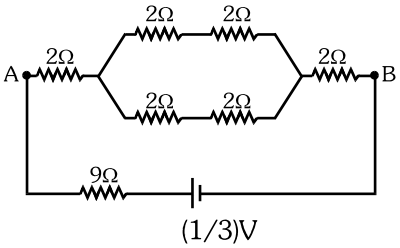
6. The electric current through ideal voltmeter is zero.

According to loop rule, E –1 × I –1 × I = 0

$\Rightarrow I = \frac{E}{2} = \frac{2}{2} = 1$ A

Reading of the voltmeter = $V_A - V_B$ = [1 × I] = [1 × 1] = 1 V

7. For maximum power, external resistance is equal to internal resistance. Therefore, 2R = 4 or R =2 Ω

8. (a) 

$R_{net} = 2 + \frac{4 \times 4}{4+4} + 2 + 9 = 15\ \Omega$

(b) Total current flow in circuit = $\frac{1/3}{15} = \frac{1}{45}$ A

Apply KVL

$V_A - 2 \times \frac{1}{45} - 4 \times \frac{1}{2 \times 45} - 2 \times \frac{1}{45} = V_B$

$V_A - V_B = \frac{2}{45} + \frac{2}{45} + \frac{2}{45} = \frac{6}{45} = \frac{2}{15}$

= 0.133 Volts

or

V_{AB} = (I × R from A to B) = $\frac{1}{45} \times (6) = \frac{6}{45}$

$= \frac{2}{15} = 0.133$ V

9. $\because \quad i_1 = \frac{R_2}{R_1 + R_2} i$

Required current = $\frac{(25+5)}{(25+5)+(10+2)} \times 1.4$

$= \left(\frac{30}{42}\right)(1.4) = 1$ A

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10. From the first balance point

$\frac{R}{S}=\frac{40}{100-40}=\frac{40}{60}$.

If 12Ω is connected in parallel with S then

$S_{eq}=\frac{12S}{12+S}$

so from new balance conditions $\frac{R}{S_{eq}}=\frac{64}{36}$

$\Rightarrow \frac{R(S+12)}{12S}=\frac{64}{36} \Rightarrow \left(\frac{40}{60}\right)\left(\frac{S+12}{12}\right)=\frac{64}{36}$

$\Rightarrow S + 12 = 32 \Rightarrow S = 20\Omega$

So $R = \left(\frac{2}{3}\right)(20) = \frac{40}{3}\Omega$

11. (a) Balanced Wheat stone Bridge

$\Rightarrow \frac{P}{Q}=\frac{R}{S}$

(b) From B to D

$\Rightarrow \frac{P}{Q}<\frac{R}{S}$ so $V_B > V_D$

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Electrostatics

SOLUTIONS

Beginner's Box - 1

1. $Q = ne = 10\%$ of $5 \times 10^{21} \times 1.6 \times 10^{-19}$

$Q = 80C$

2. Number of moles in 100 g of $CO_2 = \frac{100}{44}$ mole

and Number of electrons in one mole of $CO_2 = 22\ N_A$

so Number of electron in 100 g of CO_2

$= \frac{100}{44} \times 22\ N_A = 50\ N_A = 50 \times 6.023 \times 10^{23}$

$= 3 \times 10^{25}$ electron.

3. $N = \frac{Q}{e}$, if N is not an integer then Q is not possible.

4. Q = Number of ion × charge of each ion

= (1 billion × 2) × (+e) (1 billion = 10^9)

$= 2 \times 10^9 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-10}$ C

5. Charge Q = Ne

$\Rightarrow N = \frac{Q}{e} = \frac{3.5\times10^{-9}}{1.6\times10^{-19}} = 2.18 \times 10^{10}$

6. For proton $\Rightarrow$ u, u, d and

For neutron $\Rightarrow$ d, d, u

7. (a) $\because$ Q = Ne

$\therefore \quad N = \frac{Q}{e} = \frac{3.2\times10^{-7}}{1.6\times10^{-19}} = 2\times10^{12}$

(b) Mass gained = Nm_e

$= 2 \times 10^{12} \times 9.1 \times 10^{-31}$

$= 18.2 \times 10^{-19}$ kg

8. A will have positive charge and B will have negative charge.

Beginner's Box - 2

1. $|\vec{F}| = \frac{kq(2q)}{r^2} = \frac{2kq^2}{r^2}$

Charges on spheres after contact

$= \frac{q-2q}{2} = -\frac{q}{2}$

$|\vec{F}_{new}| = \frac{K\left(\frac{-q}{2}\right)\left(-\frac{q}{2}\right)}{r^2}$

2. $F = \frac{kq^2}{r^2} \Rightarrow q = r\sqrt{\frac{F}{k}} = 5\times10^{-10}\sqrt{\frac{4\times10^{-9}}{9\times10^9}}$

$\Rightarrow q = 5 \times 10^{-10} \times \frac{2}{3} \times 10^{-9}$

$\Rightarrow q = \frac{10}{3}\times10^{-19} \Rightarrow n = \frac{q}{e} \simeq 2$

3. In equilibrium, forces are in balancing condition $F_g = F_e$

$\Rightarrow \frac{GM^2}{d^2} = \frac{KQ^2}{d^2}$

where d is the distance between charges

$\Rightarrow \frac{Q^2}{M^2} = \frac{G}{K}$

$\frac{Q}{M} = \sqrt{\frac{G}{K}} = \sqrt{\frac{6.6\times10^{-11}}{9\times10^9}} = 0.85 \times 10^{-10}$

4. $F \propto \frac{1}{r^2}$

So $\frac{F_1}{F_2} = \frac{r_2^2}{r_1^2} \Rightarrow \frac{100}{F_2} = \frac{\left(\frac{3}{2}r\right)^2}{r^2}$

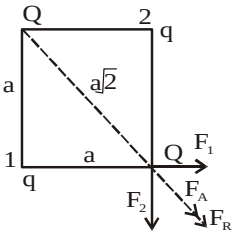
[Here $r_2 = r + 50\%r = r + r/2 = 3r/2$]

$\Rightarrow \quad F_2 = \frac{400}{9}$ N = 44.4N

5. $q = r\sqrt{\frac{F}{k}} = 4 \times \sqrt{\frac{3.6}{9\times10^9}} = 4 \times \frac{6}{3\times10^5} = 8\times10^{-5}C$

$n = \frac{q}{e} = 5 \times 10^{14}$

6.



If Net force on Q is zero

$\vec{F}_{net} = \vec{F}_R + \vec{F}_A = \vec{0}$ where F_A = Resultant of F_1 and $F_2 = \sqrt{2}\frac{kQq}{a^2}$

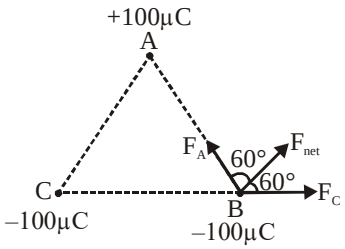
$$\Rightarrow \frac{kQ\ Q}{(\sqrt{2}a)^2}+\sqrt{2}\frac{kQ\ q}{a^2}=0$$

$$\Rightarrow \frac{kQ^2}{2a^2}+\frac{k\sqrt{2}Qq}{a^2}=0 \Rightarrow \frac{Q}{2}+\sqrt{2}q=0$$

$$\Rightarrow \frac{Q}{2}=-\sqrt{2}q$$

so $q=-\frac{Q}{2\sqrt{2}}=-\frac{\sqrt{2}}{2\sqrt{2}}=\frac{1}{2}-\mu C \quad \left[Q=\sqrt{2}\mu C\right]$

7.



$F_{net} = 2F\cos\frac{\theta}{2} = 2F\cos\left(\frac{120°}{2}\right)$

$= F = \frac{kQ^2}{r^2} \qquad \because (F_A = F_C = F)$

$$=\frac{9\times10^9(100\times10^{-6})\times(100\times10^{-6})}{(10\times10^{-2})^2}$$

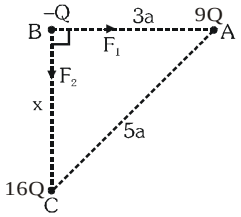
$= 9 \times 10^3$ N parallel to CA

8. Repulsive force $F = \frac{kQ^2}{r^2} = mg$

$$\Rightarrow \frac{9\times10^9\times(1.6\times10^{-19})^2}{d^2} = 1.7 \times 10^{-27} \times 9.8$$

$$\Rightarrow d = \sqrt{\frac{9\times10^9\times(1.6\times10^{-19})^2}{1.7\times10^{-27}\times9.8}} = 0.118\ m$$

9.



$x = \sqrt{(5a)^2-(3a)^2} = 4a$

$$F_1 = \frac{kQ9Q}{(3a)^2} \text{ and } F_2 = \frac{kQ16Q}{(4a)^2}$$

$$F_1 = F_2 = \frac{kQ^2}{a^2}$$

$$F_{net} = \frac{\sqrt{2}kQ^2}{a^2}$$

10. For circular motion of Q, $F_{CP} = F_e$

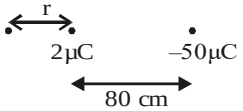
$$\Rightarrow \frac{mv^2}{R}=\frac{kQq}{R^2} \Rightarrow v=\sqrt{\frac{kQq}{mR}}$$

$$\Rightarrow \text{Time period } T = \frac{2\pi R}{v}=\frac{2\pi R\sqrt{mR}}{\sqrt{kQq}}$$

$$\Rightarrow T^2 = \frac{4\pi^2R^2(mR)4\pi\epsilon_0}{(Qq)}=\frac{16\pi^3\epsilon_0 mR^3}{Qq}$$

Beginner's Box - 3

1. For null point



Net intensity = 0

So, $\frac{kQ_1}{r_1^2}=\frac{kQ_2}{r_2^2}$

$$\frac{Q_1}{Q_2}=\frac{r_1^2}{r_2^2}\Rightarrow\frac{r_1}{r_2}=\sqrt{\frac{Q_1}{Q_2}}$$

here $r_1 = r$, $r_2 = 80 + r$

$|Q_1| = 2\ \mu C$, $|Q_2| = 50\ \mu C$

$$\Rightarrow \frac{r}{80+r}=\sqrt{\frac{2}{50}}=\frac{1}{5}$$

$\Rightarrow r = 20$ cm

2. For balancing qE = mg



$qE = 2 \times 10^{-6}g$ (1 miligram = 10^{-6} kg)

$q = \frac{2}{4} \times 10^{-5} = 5 \times 10^{-6}$ C

Nature of charge will be positive

3. $qE = mg \Rightarrow q = \frac{mg}{E}=\frac{10\times10^{-6}\times10}{100}$

$q = 1 \times 10^{-6}$

$$n = \frac{q}{e}=\frac{1\times10^{-6}}{1.6\times10^{-19}} = 6.25 \times 10^{12}$$

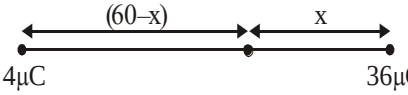
4. Electric field $\vec{E}=\frac{\vec{F}}{q}=\frac{5\times10^{-2}\hat{j}}{2\times10^{-6}}$

$= 2.5\times 10^4\ \hat{j}$ N/C

$\therefore |\vec{E}| = 2.5\times 10^4$ N/C

electric field along y-direction

5. Let at x distance from bigger charge E = 0

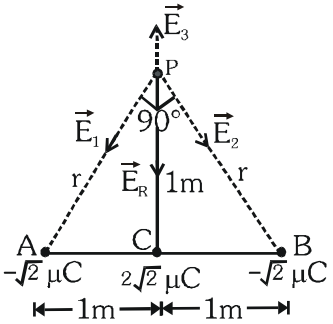


$$\frac{k(36)}{x^2}=\frac{k(4)}{(60-x)^2}=\frac{9}{x^2}=\frac{1}{(60-x)^2}$$

$$\Rightarrow \quad \frac{3}{x}=\frac{1}{60-x}\Rightarrow \ 180-3x=x$$

$\Rightarrow 180 = 4x \Rightarrow x = 45$ cm

6.



Electric field $E_1 = E_2 = \frac{kq}{r^2}$

$[\because\ r=\sqrt{2}\,m; q=\sqrt{2}\mu C; q'=2\sqrt{2}\mu C]$

$$=\frac{9\times10^9\times\sqrt{2}\times10^{-6}}{(\sqrt{2})^2}=6.36\times10^3\ N/C$$

Resultant electric field $E_R=2E_1\cos\frac{90°}{2}$

$$=2\times 6.36\times10^3\times\frac{1}{\sqrt{2}}=9\times10^3\ N/C$$

$$E_3=\frac{kq'}{(PC)^2}=\frac{9\times10^9\times2\sqrt{2}\times10^{-6}}{(1)^2}$$

$$=18\sqrt{2}\times10^3\ N/C$$

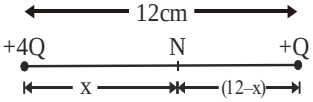
$\therefore E_{net} = E_3 - E_R = 18\sqrt{2}\times10^3 - 9\times10^3$

$= 16.46\times10^3$ N/C

Direction of net electric field is perpendicular and away from the line AB

Beginner's Box - 4

1.



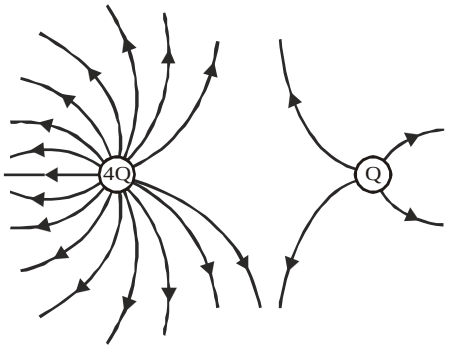
At neutral point N, the total field is zero.

Let it be at a distance x from 4Q

then $\frac{1}{4\pi\epsilon_0}\frac{4Q}{x^2}=\frac{1}{4\pi\epsilon_0}\frac{Q}{(12-x)^2}$

$$\Rightarrow\frac{(12-x)^2}{x^2}=\frac{1}{4}\Rightarrow\frac{12-x}{x}=\frac{1}{2}$$

$\Rightarrow 24 - 2x = x \Rightarrow 3x = 24 \Rightarrow x = 8$ cm



2. Surface charge density for non conducting plate $\sigma_{NC}=\frac{Q}{A}$ and

$$E_{NC}=\frac{\sigma}{2\epsilon_0}=\left(\frac{Q}{A}\right)\times\frac{1}{2\epsilon_0}=\frac{Q}{2\epsilon_0 A}=20V/m$$

For conducting plate $\sigma_C=\frac{(Q/2)}{A}=\frac{Q}{2A}$

and $E_C=\frac{\sigma}{\epsilon_0}=\left(\frac{Q}{2A}\right)\times\frac{1}{\epsilon_0}=\frac{Q}{2\epsilon_0 A}=20$ V/m

3. Net flux = Zero, because the number of lines entering the cube is same as the number of lines leaving the cube.

4. Electric flux depends on the total charge enclosed by the closed surface S. Hence flux is related with charge Q_1. Electric field at a point is the vector sum of the all electric field intensities due to all charges.

5. If the cube would have 6 faces then total flux through 6 faces = $\frac{Q}{\epsilon_0}$

Total flux through one face = $\frac{Q}{6\epsilon_0}$

Hence flux through given cube = $\frac{5}{6}\frac{Q}{\epsilon_0}$

6. Electric Flux = $\frac{q}{\epsilon_0}=\frac{2\times10^{-6}}{8.85\times10^{-12}}$

$= 2.25\times10^5\ Nm^2/C$

7. (a) From Gauss theorem $\phi=\oint_s\vec{E}.d\vec{s}=\frac{q}{\epsilon_0}$

$\Rightarrow q = \phi\times\epsilon_0$

$= (-6\times10^{-3})\times(8.85\times10^{-12})$

$= -5.31\times10^{-14}$ C

(b) Flux remains the same

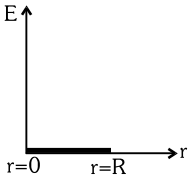
8. All i.e. q_1, q_2, q_3 & q_4

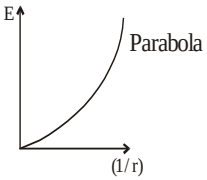
9. Yes, *Ex :* A dipole is placed in closed surface

10. (a) Zero

(b) Electric field

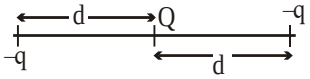
$$E=\frac{\sigma}{\epsilon_0}=\frac{17\times10^{-12}}{8.85\times10^{-12}}=1.92\ V/m$$

11. (a) 

(b) 

$$E=\frac{kq}{r^2},E\propto\left(\frac{1}{r}\right)^2,y\propto x^2$$

Beginner's Box - 5

1. 

$$U_{system}=-\frac{KqQ}{d}-\frac{KQq}{d}+\frac{Kq^2}{2d}=0$$

$$\Rightarrow\frac{2kQq}{d}=\frac{kq^2}{2d}\Rightarrow\frac{Q}{q}=\frac{1}{4}$$

2. $W = q\Delta V$

$\therefore\ \Delta V = 0$

$\Rightarrow W = 0$ (equipotential surface)

3. $$U_i=-\frac{2kq^2}{a}+\frac{4ka^2}{a}-\frac{2kq^2}{a}=0$$

$$U_f=-\frac{2kq^2}{2a}+\frac{4ka^2}{2a}-\frac{2kq^2}{2a}=0$$

$W = \Delta U = U_f - U_i = 0$

4. $$\frac{ke^2}{10\times10^{-2}}=\frac{ke^2}{20\times10^{-2}}+2\times\frac{1}{2}\times mv^2$$

$\Rightarrow v \simeq 36\ m/s$

5. (i) Potential energy $U=\frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{r}$

$$=9\times10^9\times\frac{7\times(-2)\times10^{-12}}{0.18}=-0.7\ J$$

(ii) Work required $= -U = -(-0.7)\ J = 0.7\ J$

6. (i) Potential $V=\frac{kQ}{r}=\frac{9\times10^9\times4\times10^{-7}}{9\times10^{-2}}$

$= 4\times10^4\ V$

(ii) Work done $W = q(V_f - V_i)$

$= 2\times10^{-9}(4\times10^4 - 0) = 8\times10^{-5}\ J$

$(V_i = V_\infty = 0)$

No, it is not depending on the path.

7. (a) Potential at centre of hexagon is

$$V=6\times\frac{kq}{a}=\frac{6\times9\times10^9\times5\times10^{-6}}{0.1}\ volt$$

$= 27\times10^5\ V$

8. $$\Delta V = KQ\left[\frac{1}{r_1}-\frac{1}{r_2}\right]$$

$$=9\times10^9\times5.5\times10^{-6}\left[\frac{1}{0.2}-\frac{1}{0.1}\right]$$

$= 2.475\times10^5$ volts

The point 20 cm away from charge is at higher potential

9. (i) $V=\frac{KQ}{r}\Rightarrow r=\frac{KQ}{V}=0.54m$

(ii) If r becomes four times then

$$V_{new}=\frac{V}{4}=2.5KV$$

$\Delta V = 7.5$ kV, Decrease

10. $$V=\frac{KQ_1}{r}+\frac{KQ_2}{r}+\frac{KQ_3}{r}$$

where r is distance of centre from vertex

$r = 2/3\ (20\sin60°)\times10^{-2}\ m$

$$\Rightarrow\quad r=\frac{0.2}{\sqrt{3}}m$$

$$So\ V=\frac{K\times4\times10^{-6}}{\frac{0.2}{\sqrt{3}}}=3.1\times10^5\ V$$

11. Potential at centre $V=\frac{k(Q_1+Q_2+Q_3+Q_4)}{r}$

$$r=\frac{0.1}{\sqrt{2}}m$$

$$V=\frac{k(-30\times10^{-6})}{\frac{0.1}{\sqrt{2}}}=-38.88\times10^5\ V$$

$\simeq -3.8\times10^6\ V$

12. Electric field $\vec{E}=-\frac{\partial V}{\partial x}\hat{i}-\frac{\partial V}{\partial y}\hat{j}-\frac{\partial V}{\partial z}\hat{k}$

$$=-(2x)\hat{i}-(-2y)\hat{j}-2\hat{k}=2x\,\hat{i}+2y\,\hat{j}-2\hat{k}$$

At point (1,1,1)

$$\vec{E}_P=-(2\times1)\hat{i}+(2\times1)\hat{j}-2\hat{k}$$

$$=-2\hat{i}+2\hat{j}-2\hat{k}$$

13. $\frac{kQ}{R} = V$

$\Rightarrow \frac{9\times10^{9}\times3.5\times10^{-6}}{R} = 2.5 \times 10^{3}$

$\Rightarrow R = 12.6$ m

14. (i) $\Delta V = E \times d = 3 \times 10^{4}$ volt

decreases in direction of electric field lines and vice versa.

(ii) $W = q\Delta V \therefore \Delta V = 0 \Rightarrow W = 0$

15. (a) $V_P > V_Q$ so $V_P - V_Q$ is positive

$V_B > V_A$ so $V_B - V_A$ is positive

(b) $U_Q = (-q)V_Q$, $U_P = (-q)V_P$

$U_Q - U_P = q(V_Q - V_P) \Rightarrow$ positive

similarly $U_A - U_B = -q(V_A - V_B)$

$\Rightarrow$ positive

(c) $W_{agent} = q(V_f - V_i) = q(V_P - V_Q)$ is positive so

$W_{field} = -W_{by} \Rightarrow W_{by}$ field is negative

(d) $W_{agent} = (-q)(V_A - V_B) \Rightarrow$ is positive

(e) Decreases

Beginner's Box - 6

1. $\tau_{max} = PE = 3.4 \times 10^{-30} \times 2.5 \times 10^{4}$

$= 8.5 \times 10^{-26}$ N-m

2. $\because$ Potential energy $= -PE\cos\theta$

for minimum potential energy $\theta = 0°$

3. $\tau = PE \sin\theta = q(d)E \sin\theta$

$q = \frac{\tau}{dE\sin\theta} = \frac{8\times\sqrt{3}}{4\times10^{-2}\times4\times10^{5}\sin60} = 10^{-3}C$

4. Since particle (1) and (2) are deflected opposite to E so particle (1) and (2) are negatively charged and (3) is positively charged.

The displacement $y \propto \left(\frac{Q}{m}\right)$ therefore particle (1) has highest charge to mass ratio.

5. Time taken by electron to travel 1 cm,

$t = \frac{x}{v} = \frac{10^{-2}}{10^{6}} = 10^{-8}s$

In this time it is deflected a distance antiparallel to the field which is given by –

$y = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{eE}{m}\right)t^2$

$= \frac{1}{2}\times\frac{1.6\times10^{-19}\times200}{9.1\times10^{-31}} \times 10^{-16} = 0.176$ cm

6. Total charge $= 2.5 \times 10^{-7} - 2.5 \times 10^{-7} = 0$

Dipole moment $\vec{p} = (qd)\left(-\hat{k}\right)$

$= -(2.5\times10^{-7} \times 30 \times 10^{-2})\,\hat{k} = -7.5\times10^{-8}\hat{k}$

Beginner's Box - 7

1. For any metallic surface, electric field lines are perpendicular to the surface.

2. 1, 2, & 3 are wrong because in metallic solid sphere, there is no field inside the sphere. Option (4) is correct from given reason and also it has field lines perpendicular to the surface (as required in metallic surface)

3. Due to the induction, the opposite nature of charge is induced at near by surface.